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Artificial Intelligence and Machine learning

IS353IA

UNIT 5

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What is Cluster Analysis?

- Cluster: A collection of data objects
  - similar (or related) to one another within the same group
  - dissimilar (or unrelated) to the objects in other groups
- Cluster analysis (or *clustering*, *data segmentation*, …)
  - Finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters
- Unsupervised learning: no predefined classes (i.e., *learning by observations* vs. learning by examples: supervised)
- Typical applications
  - As a stand-alone tool to get insight into data distribution
  - As a preprocessing step for other algorithms

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Clustering for Data Understanding and Applications

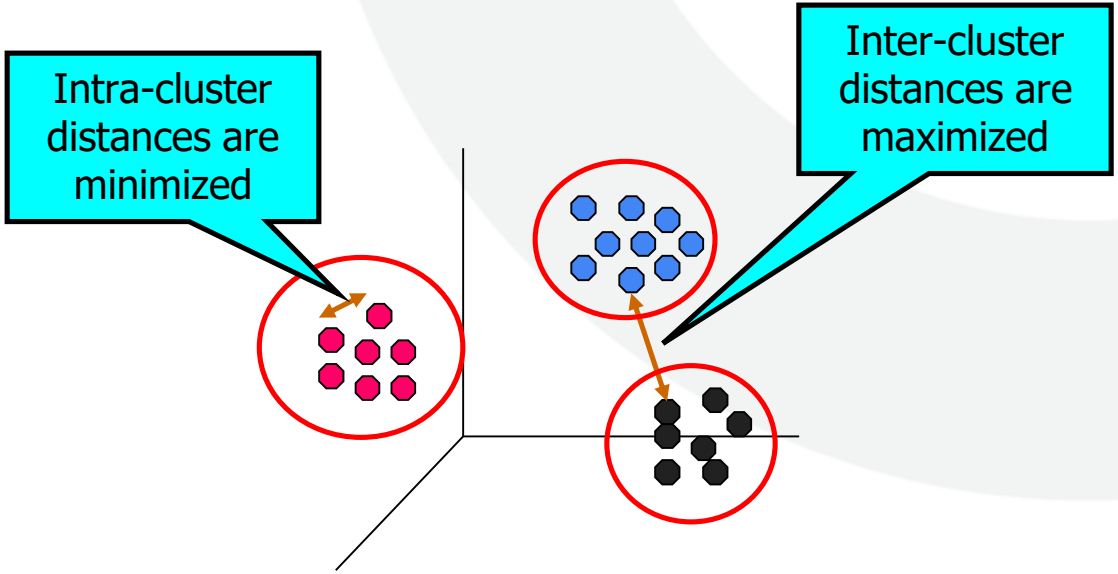
- Biology: taxonomy of living things: kingdom, phylum, class, order, family, genus and species
- Information retrieval: document clustering
- Land use: Identification of areas of similar land use in an earth observation database
- Marketing: Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs
- City-planning: Identifying groups of houses according to their house type, value, and geographical location
- Earth-quake studies: Observed earth quake epicenters should be clustered along continent faults
- Climate: understanding earth climate, find patterns of atmospheric and ocean
- Economic Science: market resarch

Clustering as a Preprocessing Tool (Utility)

- Summarization:
  - Preprocessing for regression, PCA, classification, and association analysis
- Compression:
  - Image processing: vector quantization
- Finding K-nearest Neighbors
  - Localizing search to one or a small number of clusters
- Outlier detection
  - Outliers are often viewed as those “far away” from any cluster

What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



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Quality: What Is Good Clustering?

- A good clustering method will produce high quality clusters with
  - high intra-class similarity
  - low inter-class similarity
- The quality of a clustering result depends on both the similarity measure used by the method and its implementation
- The quality of a clustering method is also measured by its ability to discover some or all of the hidden patterns

Measure the Quality of Clustering

- Dissimilarity/Similarity metric: Similarity is expressed in terms of a distance function, typically metric: $d(i, j)$
- There is a separate "quality" function that measures the "goodness" of a cluster.
- The definitions of distance functions are usually very different for interval-scaled, boolean, categorical, ordinal ratio, and vector variables.
- Weights should be associated with different variables based on applications and data semantics.
- It is hard to define "similar enough" or "good enough"
  - the answer is typically highly subjective.

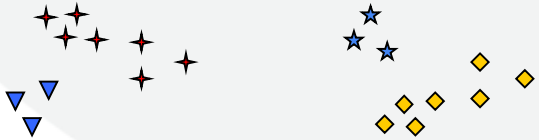
Data Type	Distance / Similarity Measure
Interval-scaled (numeric)	Euclidean, Manhattan
Boolean	Hamming distance
Categorical	Simple matching, Jaccard
Ordinal	Rank-based distance
Ratio	Normalized distance
Vector (text, images)	Cosine similarity

Notion of a Cluster can be Ambiguous

How many clusters?

Six Clusters

Two Clusters



Four Clusters

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TYPES OF DATA IN CLUSTER ANALYSIS

Data Structures

Data matrix(two modes)

A **data matrix** represents the **original dataset**.

Meaning of symbols

- **Rows (n)** → Objects / data points
- **Columns (p)** → Features / attributes
- xij → value of **j^th^ feature** of **i^th^ object**

Example:

- Rows → Students
- Columns → Marks in Math, Physics, CS

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

Dissimilarity matrix(one mode)

A **dissimilarity matrix** stores **pairwise distances** between objects.

Key points

- Matrix size → **n × n**
- Diagonal elements → **0** (distance of an object to itself)
- d(i,j)→ distance between object *i* and object *j*

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \cdots & \cdots & 0 \end{bmatrix}$$

Type of data in clustering analysis

- Interval-scaled variables
- Binary variables
- Nominal, ordinal, and ratio variables
- Variables of mixed types

Inter Values

What are interval-valued variables?

•Numeric variables where **differences matter**

•Example: height, weight, temperature, marks

Why standardize data?

Different features may have **different scales**.

•Height → 150–180

•Weight → 40–90

Without standardization, **large-scale features dominate distance**.

Interval-valued variables

- Standardize data
  - Calculate the mean absolute deviation:

$$s_f = \frac{1}{n}(|x_{1f} - m_f| + |x_{2f} - m_f| + \ldots + |x_{nf} - m_f|)$$

$$m_f = \frac{1}{n}(x_{1f} + x_{2f} + \ldots + x_{nf}).$$

where
- mf = mean of feature f
- Measures **average spread** from mean

  - Calculate the standardized measurement (*z-score*)

$$z_{if} = \frac{x_{if} - m_f}{s_f}$$

- Converts all values to **same scale**
- Mean ≈ 0, spread ≈ 1

- Using mean absolute deviation is more robust than using standard deviation

Less affected by outliers

More **robust** for real data

Binary Variables

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A contingency table for binary data(A **contingency table** is a **2 × 2 table** used to **summarize how two binary objects/variables match or differ)**

		Object j		
		1	0	sum
Object i	1	a	b	$a+b$
	0	c	d	$c+d$
	sum	$a+c$	$b+d$	p

- Distance measure for symmetric binary variables: (0 and 1 are equally important, e.g., gender):

$$d(i,j)=\frac{b+c}{a+b+c+d}$$

- Distance measure for asymmetric binary variables((1 is more important than 0, e.g., disease present , (ignores **d**, joint absence)):

$$d(i,j)=\frac{b+c}{a+b+c}$$

- Jaccard coefficient (measures shared 1s, Common in text, medical, web data).

$$sim_{Jaccard}(i,j)=\frac{a}{a+b+c}$$

Dissimilarity between Binary Variables

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- Example

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

  - gender is a symmetric attribute
  - the remaining attributes are asymmetric binary
  - let the values Y and P be set to 1, and the value N be set to 0

$$d(jack, mary) = \frac{0+1}{2+0+1} = 0.33$$

$$d(jack, jim) = \frac{1+1}{1+1+1} = 0.67$$

$$d(jim, mary) = \frac{1+2}{1+1+2} = 0.75$$

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Identify variable types

- **Gender** → *symmetric binary*
  (M/F are equally important → NOT used here)
- **Fever, Cough, Test-1 to Test-4** → *asymmetric binary*
  (presence matters more than absence)

So we **ignore Gender** for distance calculation.

Convert values to binary

Rule given:

- **Y, P → 1**
- **N → 0**

Converted table (excluding Gender)

Name	Fever	Cough	T1	T2	T3	T4
Jack	1	0	1	0	0	0
Mary	1	0	1	0	1	0
Jim	1	1	0	0	0	0

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- **Distance between Jack and Mary**

Compare feature-wise:

Feature	Jack	Mary
Fever	1	1 → a
Cough	0	0 → ignored
T1	1	1 → a
T2	0	0 → ignored
T3	0	1 → c
T4	0	0 → ignored

Counts:

- a = 2
- b = 0
- c = 1

$$d(\text{Jack}, \text{Mary}) = \frac{0+1}{2+0+1} = \frac{1}{3} = 0.33$$

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Distance between **Jack and Jim**

Feature	Jack	Jim
Fever	1	1 → a
Cough	0	1 → c
T1	1	0 → b
T2	0	0 → ignored
T3	0	0 → ignored
T4	0	0 → ignored

Counts:

- a = 1
- b = 1
- c = 1

$$d(\text{Jack, Jim}) = \frac{1+1}{1+1+1} = \frac{2}{3} = 0.67$$

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Distance between **Jim and Mary**

Feature	Jim	Mary
Fever	1	1 → a
Cough	1	0 → b
T1	0	1 → c
T2	0	0 → ignored
T3	0	1 → c
T4	0	0 → ignored

Counts:

- a = 1
- b = 1
- c = 2

$$d(\text{Jim, Mary}) = \frac{1+2}{1+1+2} = \frac{3}{4} = 0.75$$

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Pair	Distance
Jack – Mary	**0.33** (most similar)
Jack – Jim	**0.67**
Jim – Mary	**0.75** (least similar)

This problem demonstrates that choosing the correct binary distance measure based on data semantics is crucial, and asymmetric binary dissimilarity effectively captures similarity when the presence of attributes is more important than absence.

Nominal Variables

What are nominal variables?
•Categorical, **no order**
•Example: red, blue, green

- A generalization of the binary variable in that it can take more than 2 states, e.g., red, yellow, blue, green
- Method 1: Simple matching
  - *m*: # of matches, *p*: total # of variables

$$d(i,j) = \frac{p-m}{p}$$

- Method 2: use a large number of binary variables
- creating a new binary variable for each of the *M* nominal states (**Binary encoding**
- Convert each category into binary variables
- Then apply binary distance measures)

Ordinal Variables

What are ordinal variables?

•Order matters, differences don't

•Example: rank, grades (low/medium/high)

- An ordinal variable can be discrete or continuous
- Order is important, e.g., rank
- Can be treated like interval-scaled
  - replace x_{if} by their rank $r_{if} \in \{1, \ldots, M_f\}$
  - map the range of each variable onto [0, 1] by replacing *i*-th object in the *f*-th variable by

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

  - compute the dissimilarity using methods for interval-scaled variables

Ratio-Scaled Variables

- Ratio-scaled variable: a positive measurement on a nonlinear scale.

Although they are numeric, **ratio-scaled variables often follow a nonlinear or exponential scale**, such as: Ae^{Bt} or Ae^{-Bt}

- Methods:
  - treat them like interval-scaled variables—*not a good choice!* (why?—the scale can be distorted)
  - apply logarithmic transformation

$$y_{if} = log(x_{if})$$

  - treat them as continuous ordinal data treat their rank as interval-scaled

Variables of Mixed Types

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- A database may contain all the six types of variables
  - symmetric binary, asymmetric binary, nominal, ordinal, interval and ratio, A **single distance formula cannot be used directly** for all of them.
- One may use a weighted formula to combine their effects $d(i,j) = \frac{\sum_{f=1}^{p} \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^{p} \delta_{ij}^{(f)}}$
  - f is binary or nominal: $d_f(i,j) = \begin{cases} 0 & \text{if } x_{if} = x_{jf} \\ 1 & \text{otherwise} \end{cases}$
  - f is interval-based: use the normalized distance

$$d_f(i,j) = \frac{|x_{if} - x_{jf}|}{\text{range}_f}$$

  - f is ordinal or ratio-scaled
    - compute ranks r_{if} and
    - and treat z_{if} as interval-scaled

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

Meaning of each term

- f → feature index
- w_f → weight of feature f (importance)
- δ_f → 0 if missing, 1 otherwise
- $d_f(i,j)$ → distance computed **according to variable type**

This formula:

- Combines **different distance rules**
- Handles **missing values**
- Allows **feature importance**

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Types of clusters

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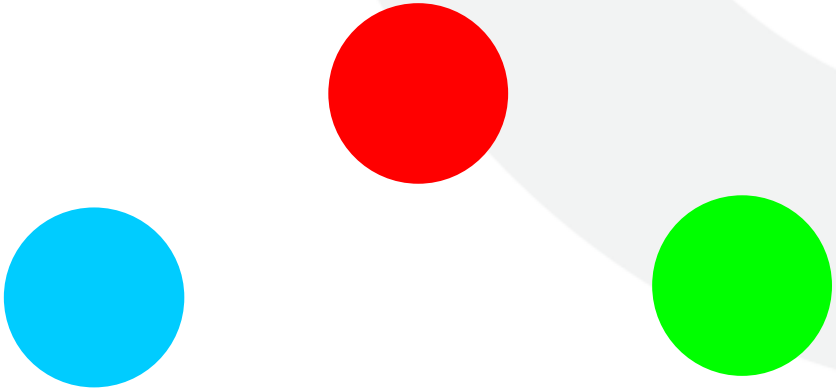
Types of Clusters

- Well-separated clusters
- Center-based clusters (Prototype clusters)
- Contiguous clusters (Graph based)
- Density-based clusters
- Shared Property or Conceptual
- Described by an Objective Function

Types of Clusters: Well-Separated

- Well-Separated Clusters:
  - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.



3 well-separated clusters

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Well-Separated Clusters

Definition: A **well-separated cluster** is a set of objects such that:

- Every object in the cluster is **closer to all other objects in the same cluster** than to **any object outside the cluster**

Key characteristics

- Clusters are **far apart**
- Very little or **no overlap**
- This is an **idealistic definition**
- Exists only when **natural clusters are clearly separated**

Shape

- Can be **any shape**, not necessarily circular

Limitation

- Rare in real-world data
- Noise and overlap usually violate this condition

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Prototype-Based (Center-Based) Clusters

Definition: A **prototype-based cluster** is a set of objects where:

- Each object is **closer to its own cluster's prototype**
- Than to the prototype of any other cluster

What is a prototype?

- **Centroid** → mean of all points (for numerical data)
- **Medoid** → most representative actual object (for categorical data)

Key characteristics

- Clusters are usually **globular (spherical)**
- Each cluster has a **central representative**

Examples

- **K-means clustering**
- **K-medoids**

Types of Clusters: Center-Based

- Center-based
  - A cluster is a set of objects such that an object in a cluster is closer (more similar) to the “center” of a cluster, than to the center of any other cluster
  - The center of a cluster is often a centroid, the average of all the points in the cluster, or a medoid, the most “representative” point of a cluster

4 center-based clusters

Graph-Based Clusters

Definition

If data is represented as a **graph:**

- Nodes → objects
- Edges → connections (similarity or proximity)

A **cluster** is defined as a **connected component:**

- Objects are connected to each other
- No connections to objects outside the cluster

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Contiguity-Based Clusters (Important type)

- Two objects are connected **only if they are within a given distance**
- A cluster forms by chaining nearby points

Advantage

- Can detect **irregular or intertwined clusters**

Major problem

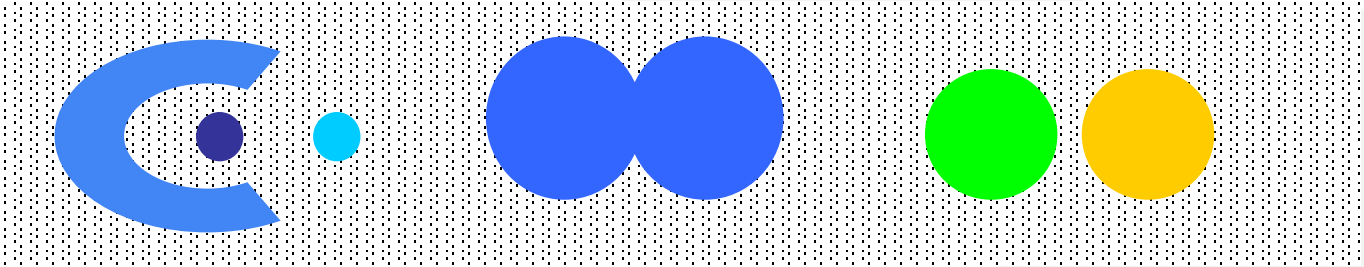
- **Noise** can create small bridges
- These bridges may **incorrectly merge separate clusters**

- Contiguous Cluster (Nearest neighbor or Transitive)
  - A cluster is a set of points such that a point in a cluster is closer (or more similar) to one or more other points in the cluster than to any point not in the cluster.

8 contiguous clusters

Types of Clusters: Density-Based

- Density-based
  - A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
  - Used when the clusters are irregular or intertwined, and when noise and outliers are present.



6 density-based clusters

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Definition

A **density-based cluster** is:

- A region of **high density of points** Surrounded by regions of **low density**

Key idea

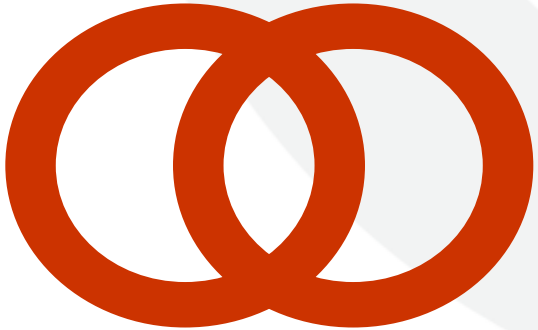
- Clusters grow where data is dense
- Sparse regions separate clusters

Comparison with contiguity-based

- Contiguity-based → fails with noise
- Density-based → robust to noise

Types of Clusters: Conceptual Clusters

- Shared Property or Conceptual Clusters
  - Finds clusters that share some common property or represent a particular concept.



2 Overlapping Circles

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Shared-Property (Conceptual) Clusters

Definition

A cluster is defined as:

A set of objects that share a **common property or concept**

This is a **very general definition**.

Conceptual Clustering

•Clusters are defined using **domain knowledge**

•Requires a **specific concept of what a cluster is**

Limitation

•Too complex for standard clustering

•Moves toward **pattern recognition**

•Hence, usually not considered in basic clustering texts

Types of Clusters: Objective Function

- In this approach, **clusters are formed by optimizing (minimizing or maximizing) a mathematical function**, called an **objective function**.
- The objective function measures how *good* a clustering is.
- Clusters Defined by an Objective Function
  - Finds clusters that minimize or maximize an objective function.
  - The algorithm tries to:

Minimize something undesirable
(e.g., distance between points inside a cluster)
OR **maximize** something desirable
(e.g., separation between different clusters)

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Enumerate all possible ways of dividing the points into clusters"

•Ideally, the algorithm would:

- •Try **every possible clustering**
- •Compute the objective function value for each
- •Pick the best one

But this is **not practical** for large datasets.

"Evaluate the goodness of each potential set of clusters"

•"Goodness" means:

- •How compact the clusters are
- •How well separated they are

•This is **quantified numerically** using the objective function.

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"Can have global or local objectives"

There are **two types of optimization goals**:

(a) Global Objective

•One single function for the **entire clustering**

•Tries to optimize clustering as a whole

Example:

•Minimize total within-cluster variance for all clusters together

(b) Local Objective

•Decisions are made **step by step**

•Each step optimizes only a **part of the clustering**

"Hierarchical clustering algorithms typically have local objectives"

•Hierarchical clustering:

- •Merges or splits clusters **one step at a time**
- •Each step makes the **best local decision**

•Does **not reconsider earlier decisions**

Hence, **local objective optimization**

"Partitional algorithms typically have global objectives"

•Partitional clustering (e.g., K-means):

- •Optimizes a **single global objective function**
- •Iteratively improves the entire clustering

Hence, **global objective optimization**

Types of Clusters: Objective Function …

- Map the clustering problem to a different domain and solve a related problem in that domain

➢ Instead of solving clustering **directly in data space**, we:
➢ Convert the problem into **another well-studied problem**
➢ Solve it there
➢ Interpret the solution back as clusters
➢ Here, clustering is mapped to a **graph problem**.

- Proximity matrix defines a weighted graph, where the nodes are the points being clustered, and the weighted edges represent the proximities between points
- Clustering is equivalent to breaking the graph into connected components, one for each cluster
- Want to minimize the edge weight between clusters and maximize the edge weight within clusters

Considerations for Cluster Analysis

- Partitioning criteria
  - Single level vs. hierarchical partitioning(Clusters inside clusters) (often, multi-level hierarchical partitioning is desirable)
- Separation of clusters
  - Exclusive (e.g., one customer belongs to only one region) vs. non-exclusive (e.g., one document may belong to more than one class)
- Similarity measure
  - Distance-based (Uses physical or mathematical distance)(e.g., Euclidian, road network, vector) vs. connectivity-based(Based on links, density, or neighborhood (e.g., density or contiguity)
- Clustering space
  - Full space(Uses **all features)** (often when low dimensional) vs. subspaces(Uses **only selected features)** (often in high-dimensional clustering)

Requirements and Challenges

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- Scalability
  - Clustering all the data instead of only on samples
- Ability to deal with different types of attributes
  - Numerical, binary, categorical, ordinal, linked, and mixture of these
- Constraint-based clustering
  - User may give inputs on constraints
  - Use domain knowledge to determine input parameters
- Interpretability and usability
- Others
  - Discovery of clusters with arbitrary shape
  - Ability to deal with noisy data
  - Incremental clustering and insensitivity to input order
  - High dimensionality

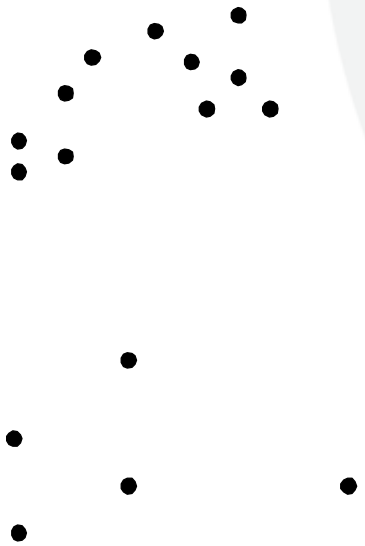
Major Clustering Approaches (I)

- Partitioning approach:
  - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
  - Typical methods: k-means, k-medoids, CLARANS
- Hierarchical approach:
  - Create a hierarchical decomposition of the set of data (or objects) using some criterion
  - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Density-based approach:
  - Based on connectivity and density functions
  - Typical methods: DBSACN, OPTICS, DenClue
- Grid-based approach:
  - based on a multiple-level granularity structure
  - Typical methods: STING, WaveCluster, CLIQUE

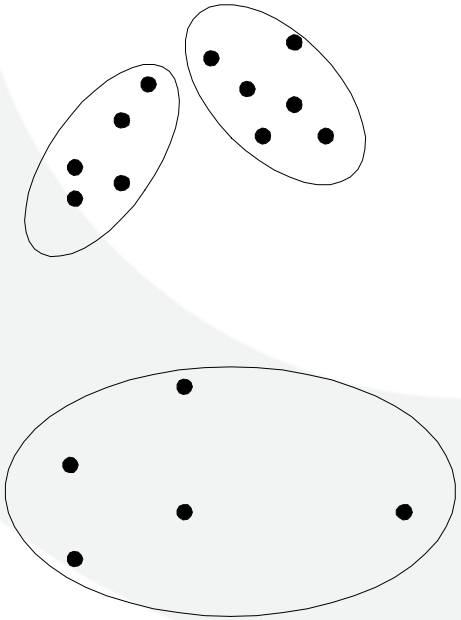
Major Clustering Approaches (II)

- Model-based:
  - A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
    - •Assume **each cluster follows a model**
      - Data → Try Model → Adjust → Best Fit → Cluster
    - •Fit the data to the **best model**
  - Typical methods: EM, SOM, COBWEB
- Frequent pattern-based:
  - Based on the analysis of frequent patterns
  - Typical methods: p-Cluster
- User-guided or constraint-based:
  - Clustering by considering user-specified or application-specific constraints
  - Typical methods: COD (obstacles), constrained clustering
- Link-based clustering:
  - Objects are often linked together in various ways
  - Massive links can be used to cluster objects: SimRank, LinkClus

Partitional Clustering

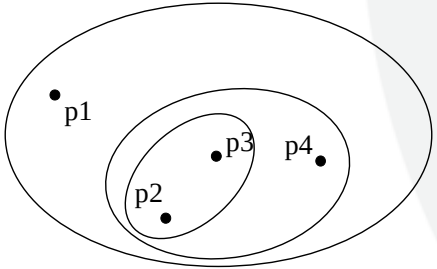


Original Points

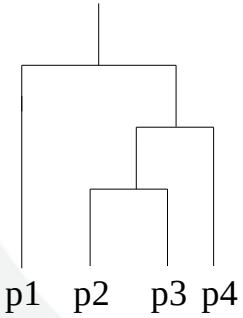


A Partitional Clustering

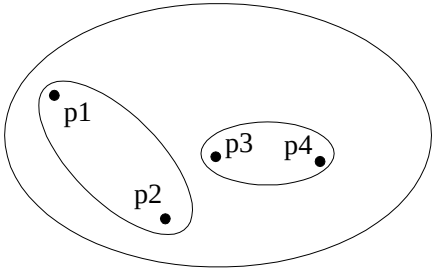
Hierarchical Clustering



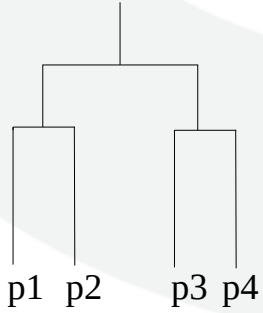
Traditional Hierarchical Clustering



Traditional Dendrogram

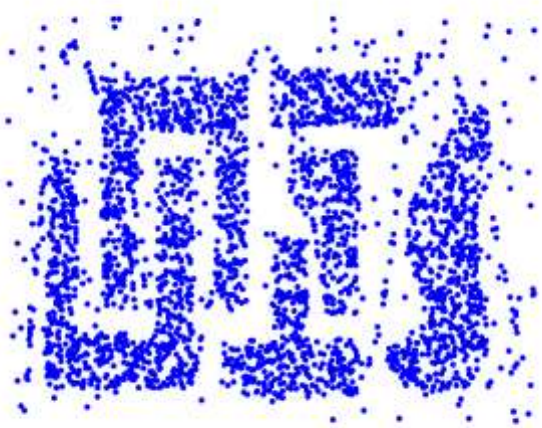


Non-traditional Hierarchical Clustering

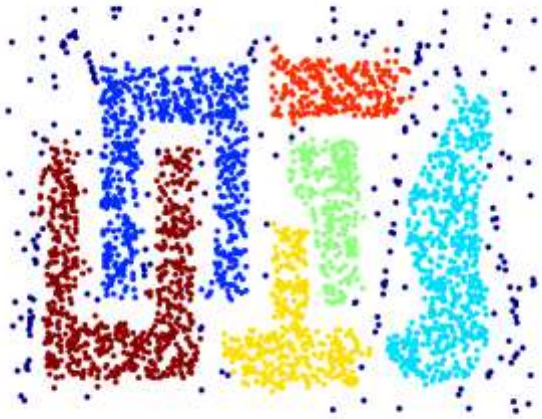


Non-traditional Dendrogram

Density based Clustering



Original Points



Clusters

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Partitional Clustering

Partitional Clustering

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- Given
  - A data set of **n objects**
  - **K the number of clusters to form**
- Organize the objects into k partitions **(k<=n) where each partition** represents a cluster
- The clusters are formed to optimize an objective partitioning criterion
  - Objects within a cluster are **similar**
  - Objects of different clusters are **dissimilar**

K-means Clustering

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- The basic algorithm is very simple
- Number of clusters, K, must be specified
- Each cluster is associated with a centroid (mean or center point)
- Each point is assigned to the cluster with the closest centroid

```
Select K points as the initial centroids.
repeat
    Form K clusters by assigning all points to the closest centroid.
    Recompute the centroid of each cluster.
until The centroids don't change
```

K-means Clustering – Details

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- Initial centroids are often chosen randomly.
  - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- 'Closeness' is measured by Euclidean distance, cosine similarity, correlation, etc.
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
  - Often the stopping condition is changed to 'Until relatively few points change clusters' or some measure of clustering doesn't change.
- Complexity is O(n * K * I * d)
  - n = number of points, K = number of clusters, I = number of iterations, d = number of attributes

K-means Clustering – Example

- Suppose that the data mining task is to cluster points into three clusters,
- where the points are
- $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, $C2(4, 9)$.
- The distance function is Euclidean distance.
- Suppose initially we assign $A1$, $B1$, and $C1$ as the center of each cluster, respectively.

K-means Clustering – Example

Initial Centroids:
A1: (2, 10)
B1: (5, 8)
C1: (1, 2)

New Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

New Centroids:
Cluster 1:
A1(2,10)
Cluster 2:
B1(8+5+7+6+4/5,4+8+5+4+9/5)
Cluster 3:
C1(2+1/2,5+2/2)

K-means Clustering – Example

Current Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

New Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-means Clustering – Example

Current Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

New Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3	9.5	6.5	5.25	1.5	3.5		
A1	2	10	1.12		6.54		6.52		1	1
A2	2	5	4.61		4.51		1.58		3	3
A3	8	4	7.43		1.95		6.52		2	2
B1	5	8	2.50		3.13		5.70		2	1
B2	7	5	6.02		0.56		5.70		2	2
B3	6	4	6.26		1.35		4.53		2	2
C1	1	2	7.76		6.39		1.58		3	3
C2	4	9	1.12		4.51		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-means Clustering – Example

Current Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3.67	9	7	4.33	1.5	3.5		
A1	2	10	1.94		7.56		6.52		1	1
A2	2	5	4.33		5.04		1.58		3	3
A3	8	4	6.62		1.05		6.52		2	2
B1	5	8	1.67		4.18		5.70		1	1
B2	7	5	5.21		0.67		5.70		2	2
B3	6	4	5.52		1.05		4.53		2	2
C1	1	2	7.49		6.44		1.58		3	3
C2	4	9	0.33		5.55		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Cluster	Points Included	Final Centroid
C1	A1,B1,C2	(3.67,9)
C2	A3,B2,B3	(7,4.33)
C3	A2,C1	(1.5,3.5)

Evaluating K-means Clusters

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- Most common measure is Sum of Squared Error (SSE)
  - For each point, the error is the distance to the nearest cluster
  - To get SSE, we square these errors and sum them.(Error = **distance from the point to its nearest cluster center)**

$$SSE = \sum_{i=1}^{K} \sum_{x \in C_i} dist^2(m_i, x)$$

  - x is a data point in cluster C_i and m_i is the representative point for cluster C_i
    - It can be shown that the **value of mi that minimizes SSE corresponds to the mean (centroid)** of the cluster.
  - Given two clusters, we can choose the one with the smallest error

- **Effect of Number of Clusters (K)**

One easy way to reduce SSE is to **increase the number of clusters (K)**.

However, **a good clustering with smaller K can have lower SSE than a poor clustering with a larger K**.

Two different K-means Clusterings

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Optimal Clustering

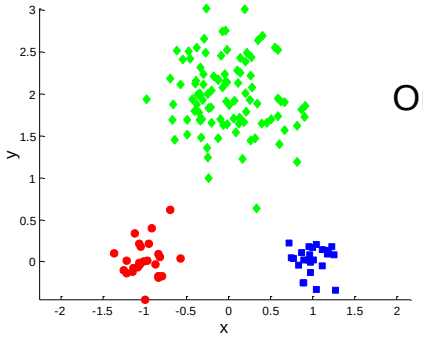
•Clusters are **compact and well separated**

•Each point is assigned to the **nearest cluster mean**

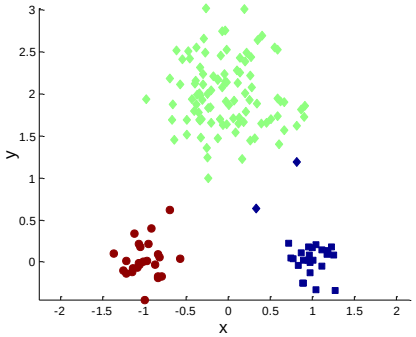
•The **Sum of Squared Error (SSE) is minimum**

•This represents a **good local minimum**

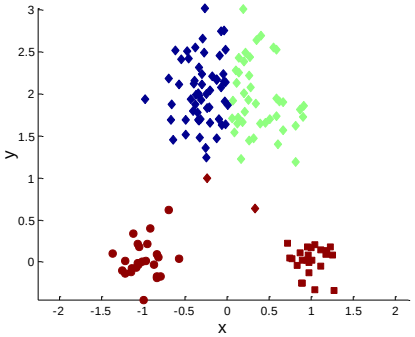
•This is the **desired result** of K-means.



Original Points



Optimal Clustering



Sub-optimal Clustering

Sub-optimal Clustering

•Some points are assigned to **less appropriate clusters**

•Cluster centers are **not ideally placed**

•**SSE is higher** than in optimal clustering

•This happens due to **different initial centroids**

K-means **can converge to a sub-optimal solution**.

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Importance of Choosing Initial Centroids

•The **final clustering depends on the initial choice of cluster centroids**.

•Different initial centroids can lead to:

- Different cluster assignments
- Different SSE values

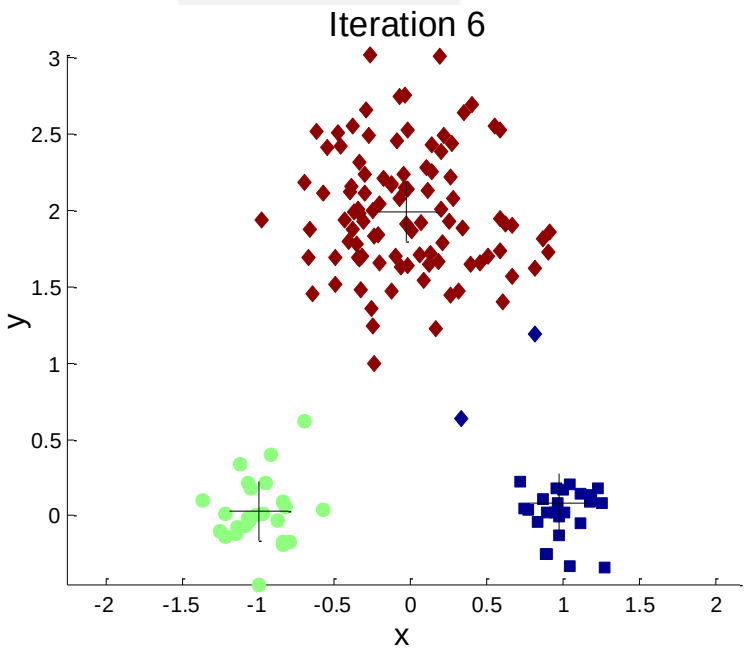
•As a result, K-means **does not guarantee a globally optimal solution**.

•It is possible for K-means to converge to a **suboptimal clustering**, even when the data has a clear natural grouping.

•Because of this sensitivity, it is common to:

- Run K-means **multiple times with different initial centroids**
- Choose the clustering with the **lowest SSE**

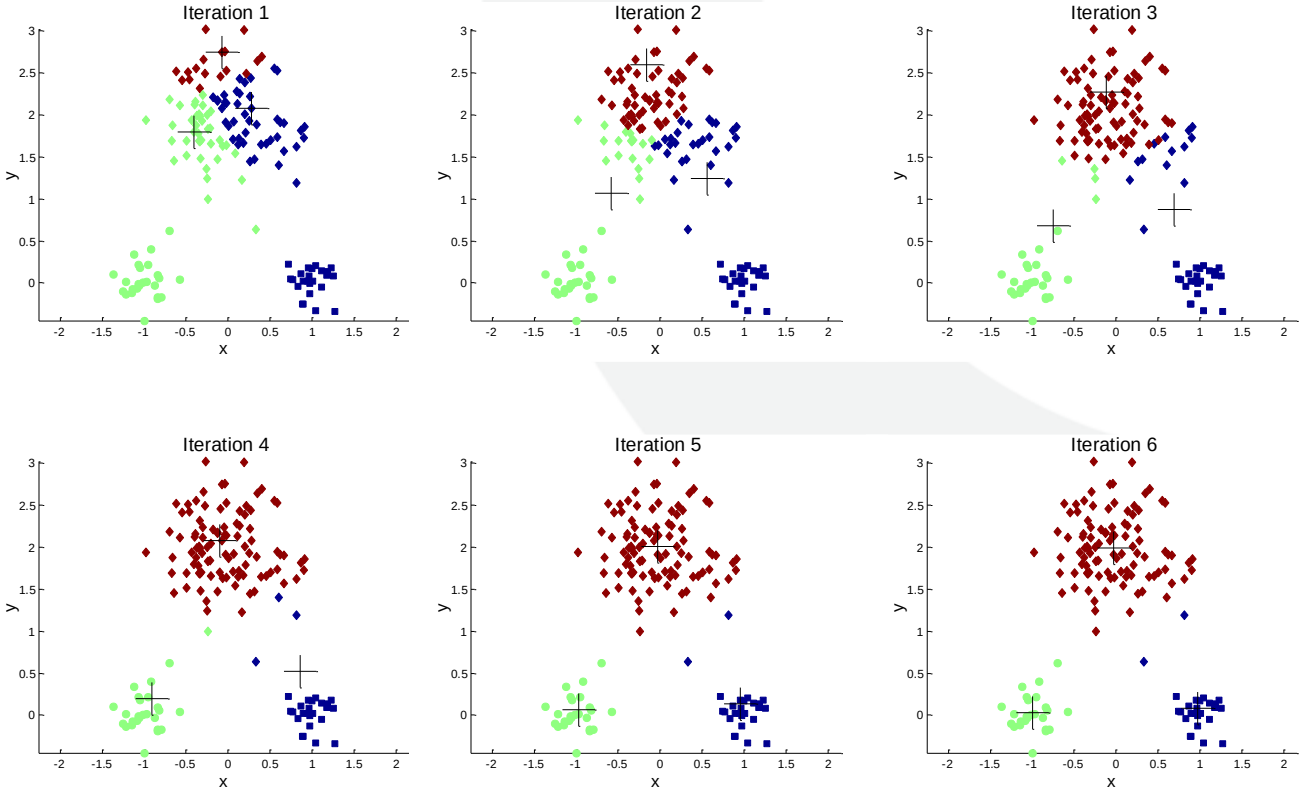
Importance of Choosing Initial Centroids (Case i)



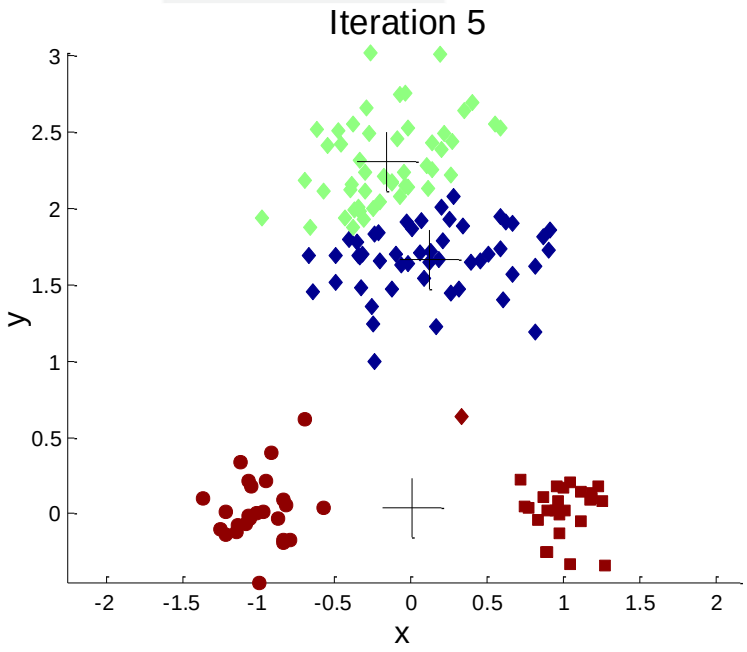
Importance of Choosing Initial Centroids (Case i)

Good Initialization

- Initial centroids are well chosen
- Clusters converge correctly
- SSE is relatively low
- Represents a **good local minimum**



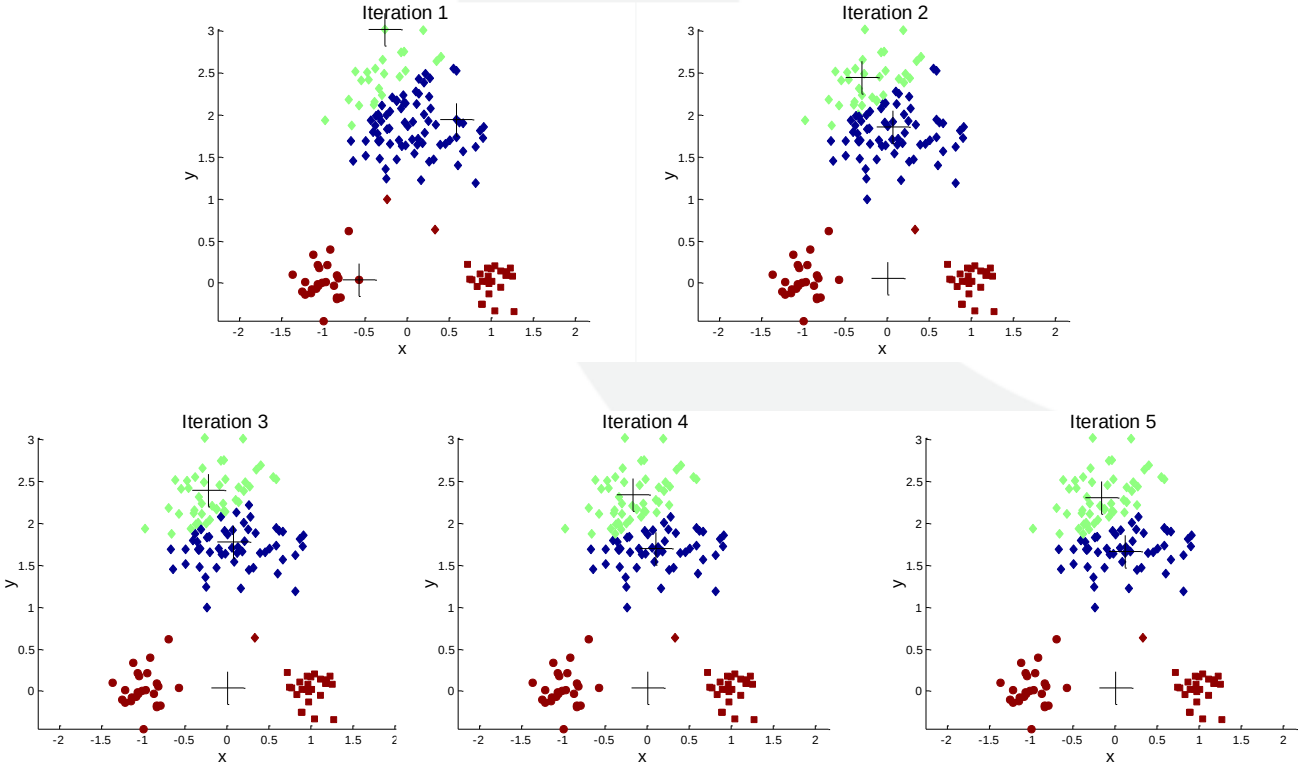
Importance of Choosing Initial Centroids (Case ii)



Importance of Choosing Initial Centroids (Case ii)

Poor Initialization

- Initial centroids are poorly chosen
- Algorithm converges to incorrect clusters
- SSE is higher
- Represents a **suboptimal local minimum**



Problems with Selecting Initial Points

Why selecting good initial centroids in K-means is difficult

Suppose there are **K "real" clusters** in the data.
K-means starts by **randomly selecting K initial centroids**.
The probability that **each initial centroid comes from a different real cluster** is **very small**, especially when:
•The number of clusters **K is large**
•The clusters are of **equal size**

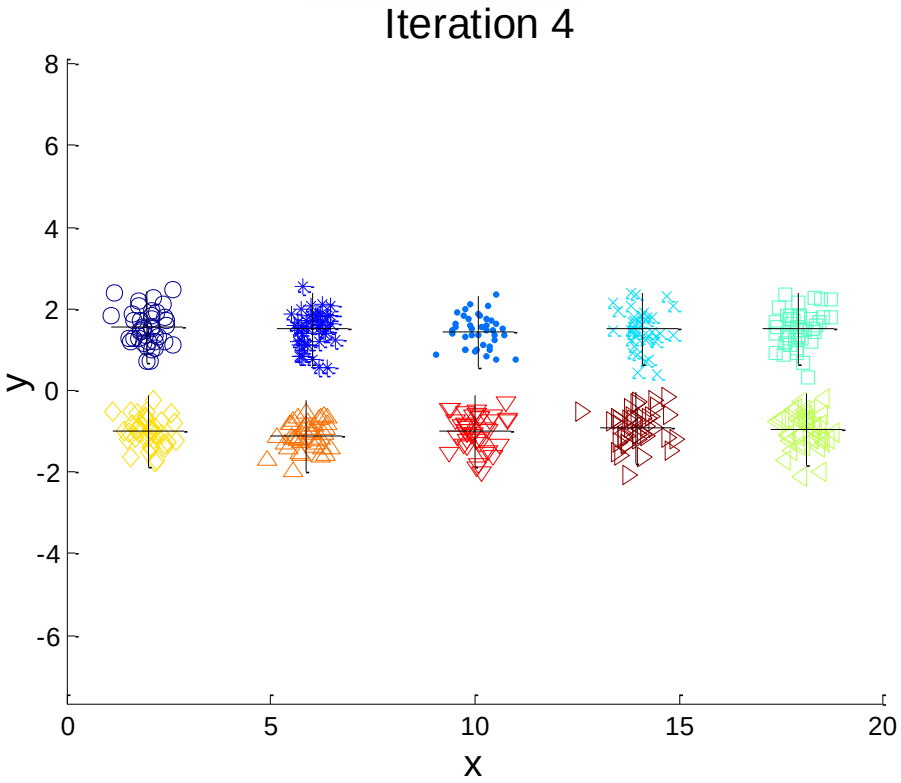
- For example, if K = 10, then probability = $10!/10^{10} = 0.00036$
- Sometimes the initial centroids will readjust themselves in 'right' way, and sometimes they don't

$$P = \frac{\text{number of ways to select one centroid from each cluster}}{\text{number of ways to select } K \text{ centroids}} = \frac{K!n^K}{(Kn)^K} = \frac{K!}{K^K}$$

•There are **K clusters**
•Each cluster has **n points**

This shows that the probability **decreases rapidly as K increases.**

10 Clusters Example (Good Clusters)



Starting with two initial centroids in one cluster of each pair of clusters

Solutions to Initial Centroids Problem

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- Multiple Runs
  Run K-means multiple times with different random initial centroids
  Select the clustering with the smallest SSE
  This approach helps, but when K is large the probability of selecting good initial centroids is still small
- Sample and Use Hierarchical Clustering
  Apply hierarchical clustering to a sample of the data
  Use the resulting clusters to determine initial centroids for K-means
  This improves the quality of initialization compared to random selection
- Select More Than K Initial Centroids
  Select more than K candidate initial centroids
  Choose the most widely separated points among them
  Use these selected points as the initial centroids
- Post-processing
  Apply post-processing techniques after K-means has converged
  Improve cluster quality when the initial clustering is poor
- Bisecting K-means
  A variant of the K-means algorithm. Works by repeatedly splitting clusters. Less susceptible to initialization problems than standard K-means

Pre-processing and Post-processing

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- Pre-processing
  - Normalize the data
  - Eliminate outliers
- Post-processing
  - Eliminate small clusters that may represent outliers
  - Split ‘loose’ clusters, i.e., clusters with relatively high SSE
  - Merge clusters that are ‘close’ and that have relatively low SSE
  - Can use these steps during the clustering process
    - ISODATA

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Agglomerative Hierarchical Clustering

Hierarchical clustering techniques are a second important category of cluster ing methods. As with K-means, these approaches are relatively old compared to many clustering algorithms, but they still enjoy widespread use.

There are two basic approaches for generating a hierarchical clustering:

Agglomerative: Start with the points as individual clusters and, at each step, merge the closest pair of clusters. This requires defining a notion of cluster proximity.

Divisive: Start with one, all-inclusive cluster and, at each step, split a cluster until only singleton clusters of individual points remain. In this case, we need to decide which cluster to split at each step and how to do the splitting.

A hierarchical clustering is often displayed graphically using a tree-like diagram called a dendrogram

Algorithm 8.3 Basic agglomerative hierarchical clustering algorithm.

```
Compute the proximity matrix, if necessary.
repeat
    Merge the closest two clusters.
    Update the proximity matrix to reflect the proximity between the new
    cluster and the original clusters.
until Only one cluster remains.
```

Bisecting K-means

The bisecting K-means algorithm is a straightforward extension of the basic K-means algorithm that is based on a simple idea: to obtain K clusters, split the set of all points into two clusters, select one of these clusters to split, and so on, until K clusters have been produced.

Bisecting K-means creates clusters by **repeatedly splitting data into two parts**.

Instead of forming K clusters at once, it starts with all points in one cluster and keeps dividing one cluster into two until K clusters are formed.

Basic Idea

- Start with **all data points in one cluster**.
- Split the cluster into **two sub-clusters** using K-means (K = 2).
- Select one of the clusters (usually with highest variance or SSE).
- Repeat the splitting process until **K clusters** are formed.

Bisecting K-means

```
Initialize the list of clusters to contain the cluster containing all points.
repeat
    Select a cluster from the list of clusters
    for i = 1 to number_of_iterations do
        Bisect the selected cluster using basic K-means
    end for
    Add the two clusters from the bisection with the lowest SSE to the list of clusters.
until Until the list of clusters contains K clusters
```

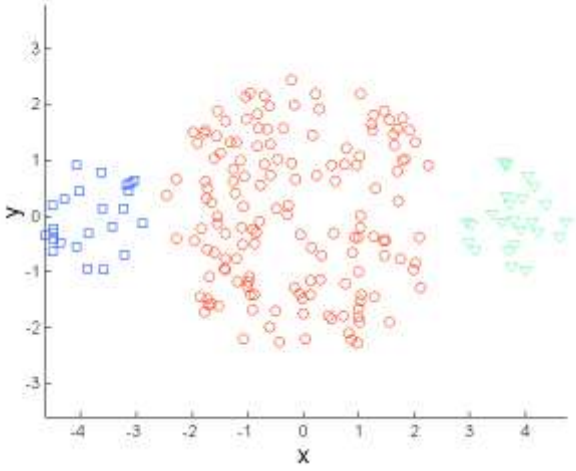
K-means and Different Types of Clusters

- K-means and its variations have a number of limitations with respect to finding different types of clusters.
- In particular, K-means has difficulty detecting the “natural” clusters, when clusters have non-spherical shapes or widely different sizes or densities.

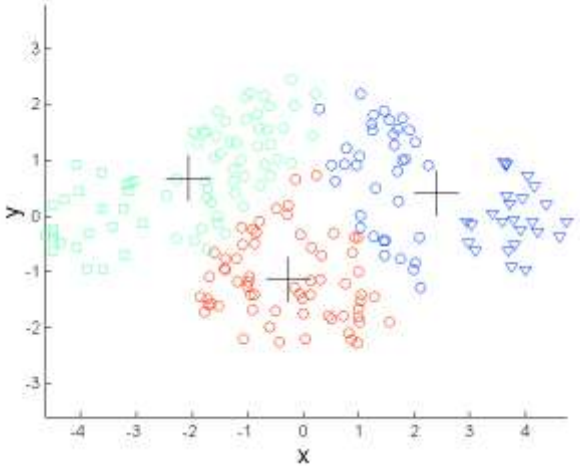
Limitations of K-means

- K-means has problems when clusters are of differing
  - Sizes
  - Densities
  - Non-globular shapes

Limitations of K-means: Differing Sizes



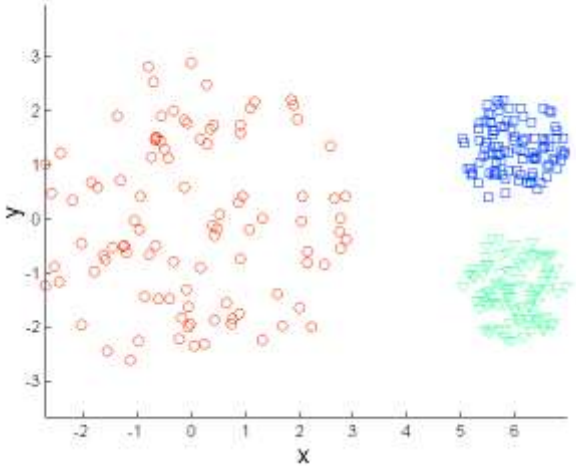
Original Points



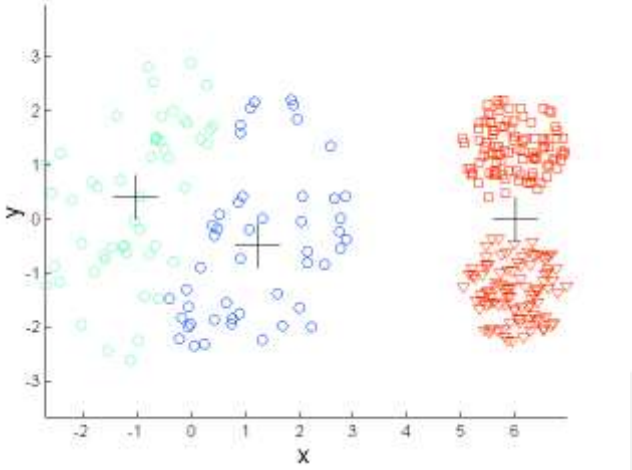
K-means (3 Clusters)

In Figure, one cluster is much larger than the other two.
Because of this size difference, K-means splits the large cluster and incorrectly merges part of it with a smaller cluster instead of finding the three natural clusters.

Limitations of K-means: Differing Density



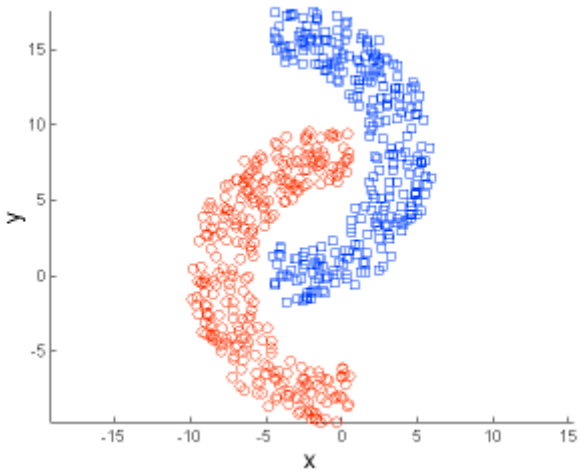
Original Points



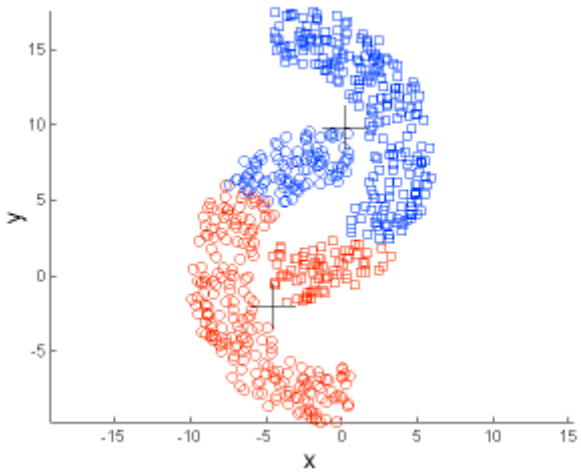
K-means (3 Clusters)

In Figure, two clusters are much denser than the third cluster.
Because of this density difference, K-means cannot correctly separate the three natural clusters.

Limitations of K-means: Non-globular Shapes



Original Points



K-means (2 Clusters)

In Figure , the natural clusters do not have a globular shape.
Because of this, K-means mixes parts of the natural clusters and forms incorrect clusters.

The problem in these cases is that the K-means objective function does not match the type of clusters present in the data.

K-means works best for
globular clusters with similar size and density or
for well-separated clusters,

However, these limitations can be overcome, in some sense, if the user is willing to accept a clustering that breaks the natural clusters into a number of subclusters.

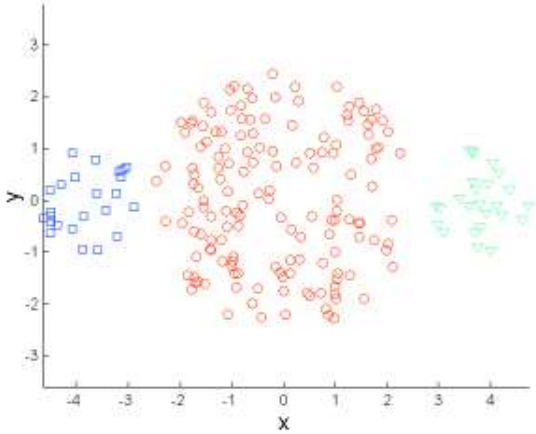
Overcoming K-means Limitations

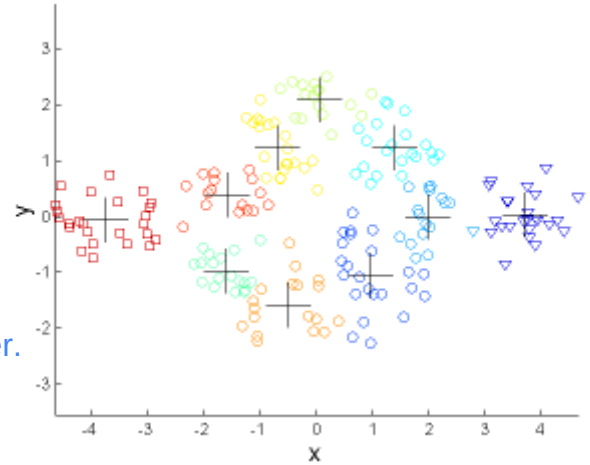
•Instead of finding two or three clusters, **n clusters are formed**

•Each natural cluster is broken into **smaller subclusters**

•Each subcluster is **pure**, meaning it contains points from only one natural cluster

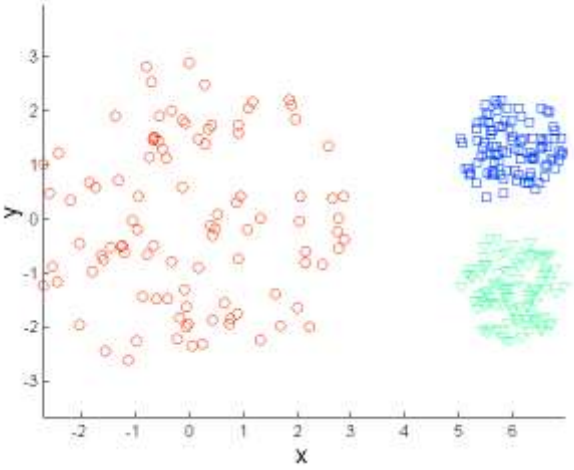
•Although natural clusters are split, their structure is still captured in parts



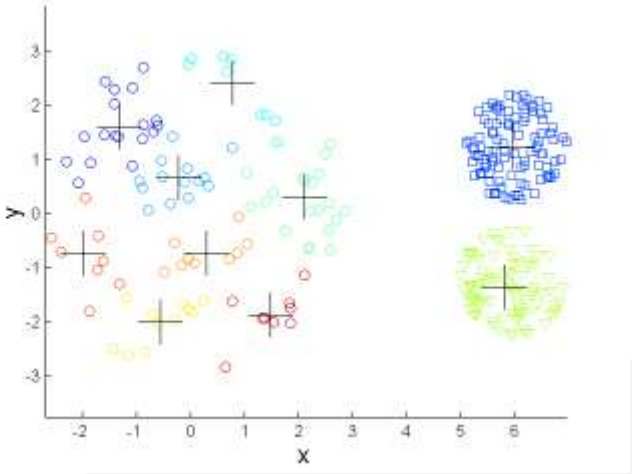


- One solution is to use many clusters.
  - Find parts of clusters, but need to put together.

Overcoming K-means Limitations



Original Points

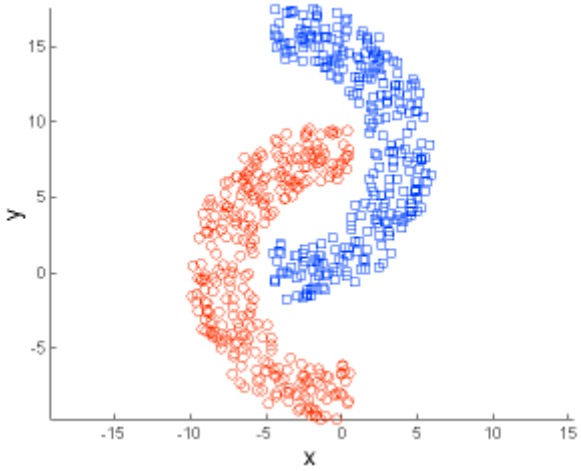


K-means

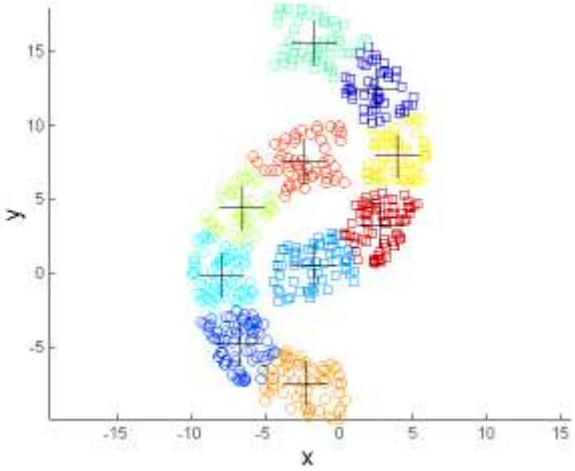
Clusters

- One solution is to use many clusters.
  - Find parts of clusters, but need to put together.

Overcoming K-means Limitations



Original Points



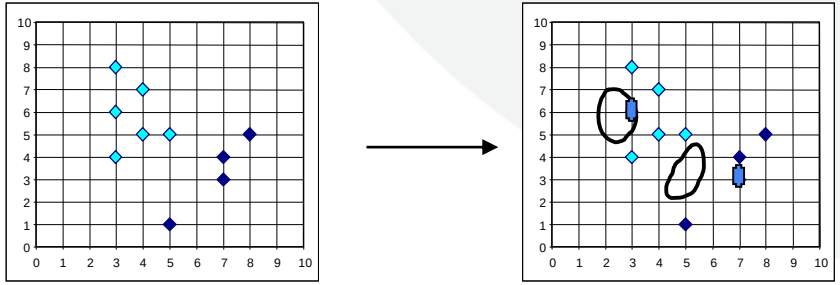
K-means

Clusters

- One solution is to use many clusters.
  - Find parts of clusters, but need to put together.

Limitations of K-means: Outlier Problem

- The k-means algorithm is sensitive to outliers !
  - Since an object with an extremely large value may substantially distort the distribution of the data.
- Solution: Instead of taking the **mean** value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster.
- A **medoid** is an **actual data point** that lies near the center of a cluster.
- Unlike the **mean**, it is **not an average** — it is one of the real objects in the data.



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Strengths and Weaknesses of K means

Advantages of K-means

- Simple and easy to implement
- Efficient for large datasets
- Variants such as bisecting K-means improve efficiency and initialization

Limitations of K-means

- Cannot handle non-globular clusters
- Performs poorly with clusters of different sizes and densities
- Sensitive to outliers

Handling Limitations

- Increasing the number of clusters produces pure subclusters
- Outlier detection and removal improve clustering resul

K-means as an Optimization Problem

K-means as an Optimization Problem

- Clustering can be treated as an optimization problem
- The objective function is usually to minimize SSE

Global vs Practical Solution

- Enumerating all possible clustering finds the global optimum
- This approach is computationally infeasible

Optimization Approach Used

- Start with an initial clustering solution
- Repeatedly update the solution to reduce SSE
- This approach is similar to gradient descent

Simplifying Assumption

- Data is assumed to be one-dimensional
- Distance is defined as $(x - y)^2$
- This simplifies the mathematical analysis

Unit 5

Cluster Evaluation-Overview, Unsupervised Cluster Evaluation Using Cohesion and Separation, Unsupervised Cluster Evaluation Using the Proximity Matrix, Determining the Correct Number of Clusters, Supervised Measures of Cluster Validity, Assessing the Significance of Cluster Validity Measures, Choosing a Cluster Validity Measure

Cluster Evaluation: hard problem

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- The quality of a clustering is very hard to evaluate because
  - We do not know the correct clusters
- Some methods are used:
  - User inspection
    - Study centroids, and spreads
    - Rules from a decision tree.
    - For text documents, one can read some documents in clusters.

Cluster evaluation: ground truth

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- We use some labeled data (for classification)
- Assumption: Each class is a cluster.
- After clustering, a confusion matrix is constructed. From the matrix, we compute various measurements, entropy, purity, precision, recall and F-score.
  - Let the classes in the data D be $C = (c_1, c_2, \ldots, c_k)$. The clustering method produces k clusters, which divides D into k disjoint subsets, $D_1, D_2, \ldots, D_k$.

Evaluation measures: Entropy

Entropy: For each cluster, we can measure its entropy as follows:

$$entropy(D_i) = -\sum_{j=1}^{k} \Pr_i(c_j) \log_2 \Pr_i(c_j), \tag{29}$$

where $\Pr_i(c_j)$ is the proportion of class c_j data points in cluster i or D_i. The total entropy of the whole clustering (which considers all clusters) is

$$entropy_{total}(D) = \sum_{i=1}^{k} \frac{|D_i|}{|D|} \times entropy(D_i) \tag{30}$$

$Pr_i(c_j)$:Proportion of class $c_j$in cluster D_i , $k = total\ num\ of\ classes$

If all points in Di belong to one class-> Pri(cj)=1-> entropy =0

If points are evenly split among classes -> entropy is maximized.

Evaluation measures: purity

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Purity: This again measures the extent that a cluster contains only one class of data. The purity of each cluster is computed with

$$purity(D_i) = \max_j(\Pr_i(c_j)) \tag{31}$$

The total purity of the whole clustering (considering all clusters) is

$$purity_{total}(D) = \sum_{i=1}^{k} \frac{|D_i|}{|D|} \times purity(D_i) \tag{32}$$

Purity- measures how well each cluster aligns with a single class.
A cluster with only 1 class has maximum purity(value=1)
A cluster with mixed classes has lower purity

An example

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Example 14: Assume we have a text collection D of 900 documents from three topics (or three classes), Science, Sports, and Politics. Each class has 300 documents. Each document in D is labeled with one of the topics (classes). We use this collection to perform clustering to find three clusters. Note that class/topic labels are not used in clustering. After clustering, we want to measure the effectiveness of the clustering algorithm.

Cluster	Science	Sports	Politics		Entropy	Purity
1	250	20	10		0.589	0.893
2	20	180	80		1.198	0.643
3	30	100	210		1.257	0.617
Total	300	300	300		1.031	0.711

An example

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Purity of a cluster = proportion of the **dominant class** in that cluster.

Cluster 1:

Dominant class: Science (250)

Purity = $\frac{250}{280} \approx 0.893$

Cluster 2:

Dominant class: Sports (180)

Purity = $\frac{180}{280} \approx 0.643$

Cluster 3:

Dominant class: Politics (210)

Purity = $\frac{210}{340} \approx 0.617$

Total Purity:

Weighted average:

$$\text{Purity}_{total} = \frac{280}{900} \cdot 0.893 + \frac{280}{900} \cdot 0.643 + \frac{340}{900} \cdot 0.617 \approx 0.711$$

An example

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Entropy of a cluster: Entropy$(D_i) = -\sum_{j=1}^{3} P\,r_i(c_j) \log_2 Pr_i(c_j)$

Where $Pr_i(c_j)$is the proportion of class $c_j$in cluster D_i.

Cluster 1:

Science: $\frac{250}{280} \approx 0.893$

Sports: $\frac{20}{280} \approx 0.071$

Politics: $\frac{10}{280} \approx 0.036$

Entropy ≈

$$-(0.893 \log_2 0.893 + 0.071 \log_2 0.071 + 0.036 \log_2 0.036) \approx 0.589$$

Cluster 2:

Sports: $\frac{180}{280} \approx 0.643$

Politics: $\frac{80}{280} \approx 0.286$

Science: $\frac{20}{280} \approx 0.071$

Entropy ≈ 1.198

An example

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Cluster 3:

Politics: $\frac{210}{340} \approx 0.617$

Sports: $\frac{100}{340} \approx 0.294$

Science: $\frac{30}{340} \approx 0.088$

Entropy ≈ 1.257

Total Entropy:

Weighted average:

$$\text{Entropy}_{total} = \frac{280}{900} \cdot 0.589 + \frac{280}{900} \cdot 1.198 + \frac{340}{900} \cdot 1.257 \approx 1.031$$

Metric	Meaning	Value
Purity	Higher is better (closer to 1)	0.711
Entropy	Lower is better (closer to 0)	1.031

A remark about ground truth evaluation

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- Commonly used to compare different clustering algorithms.
- A real-life data set for clustering has no class labels.
  - Thus although an algorithm may perform very well on some labeled data sets, no guarantee that it will perform well on the actual application data at hand.
- The fact that it performs well on some label data sets does give us some confidence of the quality of the algorithm.
- This evaluation method is said to be based on **external data** or information.

Evaluation based on internal information

- **Intra-cluster cohesion** (compactness):
  - Cohesion measures how near the data points in a cluster are to the cluster centroid.
  - Sum of squared error (SSE) is a commonly used measure.
- **Inter-cluster separation** (isolation):
  - Separation means that different cluster centroids should be far away from one another.
- In most applications, expert judgments are still the key.

Metric	Goal	Indicator of Good Clustering
Intra-cluster cohesion	Minimize within-cluster spread	Low SSE
Inter-cluster separation	Maximize distance between clusters	High centroid separation

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Indirect evaluation

- In some applications, clustering is not the primary task, but used to help perform another task.
- We can use the performance on the primary task to compare clustering methods.
- For instance, in an application, the primary task is to provide recommendations on book purchasing to online shoppers.
  - If we can cluster books according to their features, we might be able to provide better recommendations.
  - We can evaluate different clustering algorithms based on how well they help with the recommendation task.
  - Here, we assume that the recommendation can be reliably evaluated.

Assessing Clustering Tendency

- Assess if non-random structure exists in the data by measuring the probability that the data is generated by a uniform data distribution
- Test spatial randomness by statistic test: Hopkins Static
  - Given a dataset D regarded as a sample of a random variable o, determine how far away o is from being uniformly distributed in the data space
  - Sample *n* points, $p_1, \ldots, p_n$, uniformly from D. For each p_i, find its nearest neighbor in D: $x_i = min\{dist\ (p_i, v)\}$ where *v* in D
  - Sample *n* points, $q_1, \ldots, q_n$, uniformly from D. For each q_i, find its nearest neighbor in $D - \{q_i\}$: $y_i = min\{dist\ (q_i, v)\}$ where *v* in D and $v \neq q_i$
  - Calculate the Hopkins Statistic:

  - If D is uniformly distributed, $\sum x_i$ and $\sum y_i$ will be close to each other and H is close to 0.5. If D is highly skewed, H is close to 0

$$H = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i}$$

Determine the Number of Clusters

Disadvantage of K means clustering:

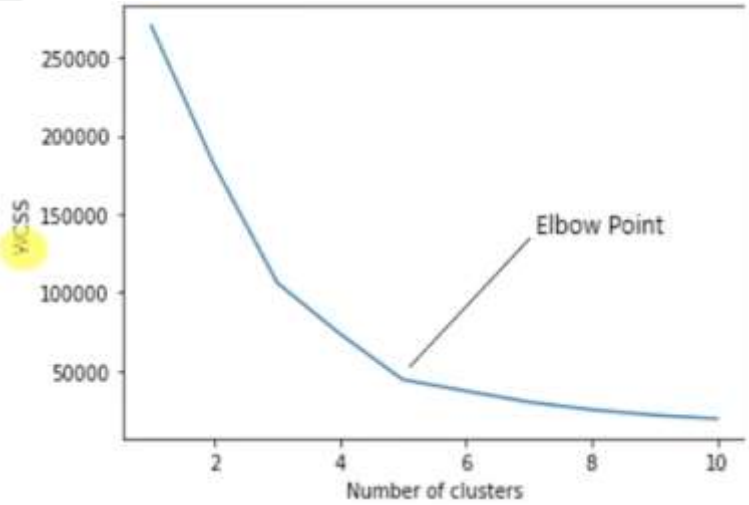
1) Need to select initial centroids for each of the cluster

2) It requires to specify the number of clusters in advance

There are two methods to select optimal number of clusters in K-means algorithm:

Elbow method

Silhouette method



- **Elbow method**
  - Use the turning point in the curve of sum of within cluster
  - variance w.r.t the # of clusters

Steps:

- Run k-means for multiple values of *k*
- Compute SSE for each
- Plot SSE vs. *k*
- Identify the elbow point- where adding more clusters doesn't reduce SSE significantly

Determine the Number of Clusters

- Empirical method
  - # of clusters ≈$\sqrt{n/2}$ for a dataset of n points
- Elbow method
  - Use the turning point in the curve of sum of within cluster variance w.r.t the # of clusters
- Cross validation method
  - Divide a given data set into *m* parts
  - Use *m* – 1 parts to obtain a clustering model
  - Use the remaining part to test the quality of the clustering
    - E.g., For each point in the test set, find the closest centroid, and use the sum of squared distance between all points in the test set and the closest centroids to measure how well the model fits the test set
  - For any k > 0, repeat it *m* times, compare the overall quality measure w.r.t. different *k's*, and find # of clusters that fits the data the best

Measuring Clustering Quality

- Two methods: extrinsic vs. intrinsic
- Extrinsic: supervised, i.e., the ground truth is available
  - Compare a clustering against the ground truth using certain clustering quality measure
  - Ex. BCubed precision and recall metrics
- Intrinsic: unsupervised, i.e., the ground truth is unavailable
  - Evaluate the goodness of a clustering by considering how well the clusters are separated, and how compact the clusters are
  - Ex. Silhouette coefficient

Measuring Clustering Quality: Extrinsic Methods

- Clustering quality measure: $Q(C, C_g)$, for a clustering C given the ground truth C_g.
- Q is good if it satisfies the following **4** essential criteria
  - Cluster homogeneity: the purer, the better
  - Cluster completeness: should assign objects belong to the same category in the ground truth to the same cluster
  - Rag bag: putting a heterogeneous object into a pure cluster should be penalized more than putting it into a *rag bag* (i.e., “miscellaneous” or “other” category)
  - Small cluster preservation: splitting a small category into pieces is more harmful than splitting a large category into pieces

Different Aspects of Cluster Validation

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1. Determining the clustering tendency of a set of data, i.e., distinguishing whether non-random structure actually exists in the data.
2. Comparing the results of a cluster analysis to externally known results, e.g., to externally given class labels.
3. Evaluating how well the results of a cluster analysis fit the data *without* reference to external information.
   - Use only the data
4. Comparing the results of two different sets of cluster analyses to determine which is better.
5. Determining the 'correct' number of clusters.

For 2, 3, and 4, we can further distinguish whether we want to evaluate the entire clustering or just individual clusters.

Measures of Cluster Validity

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- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
  - External Index: Used to measure the extent to which cluster labels match externally supplied class labels.
    - Entropy
  - Internal Index: Used to measure the goodness of a clustering structure *without* respect to external information.
    - Sum of Squared Error (SSE)
  - Relative Index: Used to compare two different clusterings or clusters.
    - Often an external or internal index is used for this function, e.g., SSE or entropy
- Sometimes these are referred to as criteria instead of indices
  - However, sometimes criterion is the general strategy and index is the numerical measure that implements the criterion.

Measuring Cluster Validity Via Correlation

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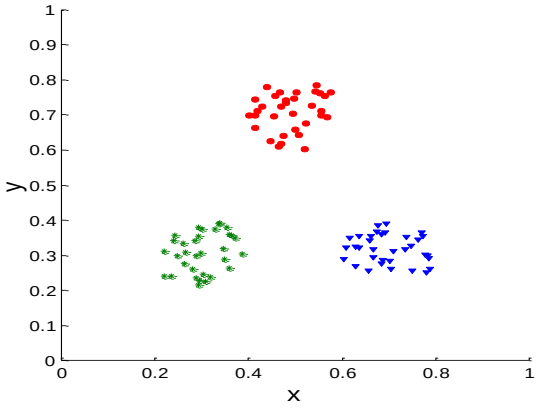
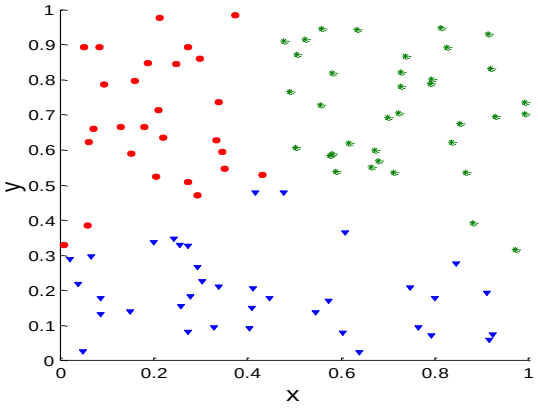
- Two matrices
  - Proximity Matrix
  - "Incidence" Matrix
    - One row and one column for each data point
    - An entry is 1 if the associated pair of points belong to the same cluster
    - An entry is 0 if the associated pair of points belongs to different clusters
- Compute the correlation between the two matrices
  - Since the matrices are symmetric, only the correlation between n(n-1) / 2 entries needs to be calculated.
- High correlation indicates that points that belong to the same cluster are close to each other.
- Not a good measure for some density or contiguity based clusters.

Measuring Cluster Validity Via Correlation

- Correlation of incidence and proximity matrices for the K-means clusterings of the following two data sets.



Corr = -0.9235



Corr = -0.5810

Using Similarity Matrix for Cluster Validation

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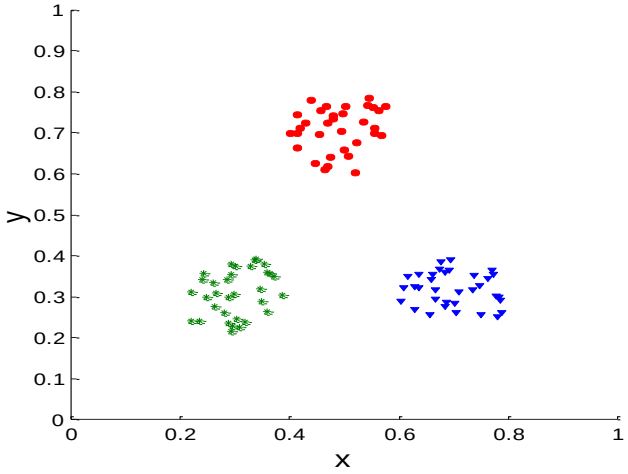
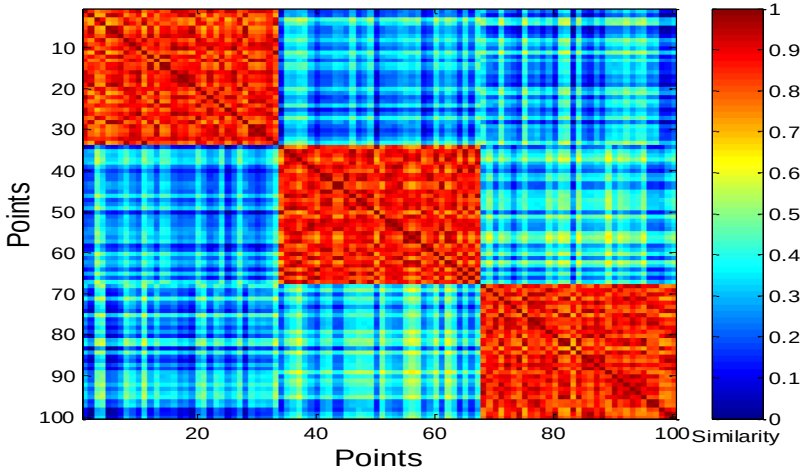
- Order the similarity matrix with respect to cluster labels and inspect visually.





Using Similarity Matrix for Cluster Validation

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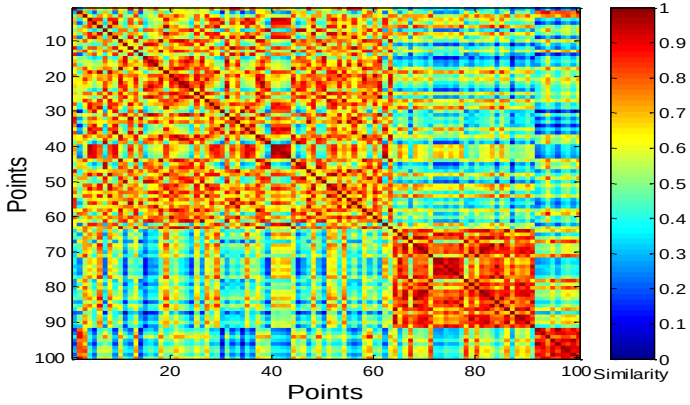
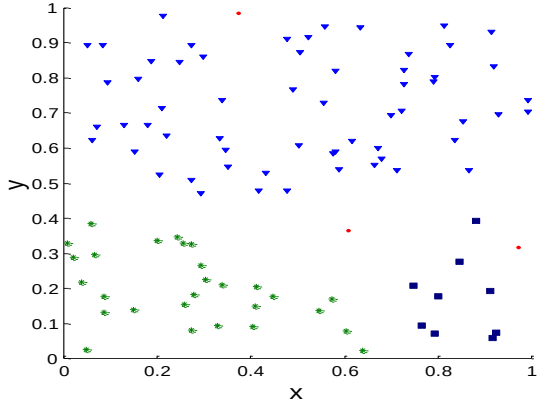
- Clusters in random data are not so crisp





DBSCAN

Using Similarity Matrix for Cluster Validation

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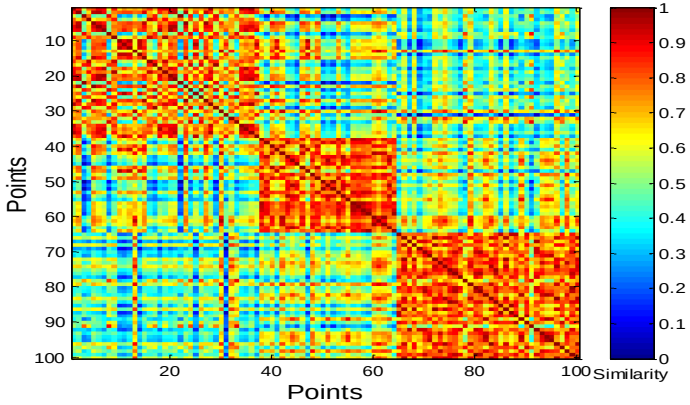
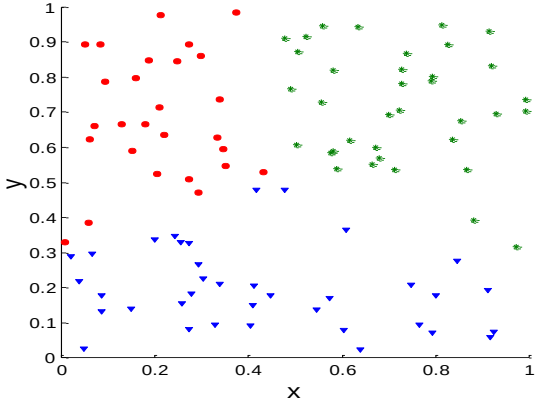
- Clusters in random data are not so crisp





K-means

Using Similarity Matrix for Cluster Validation

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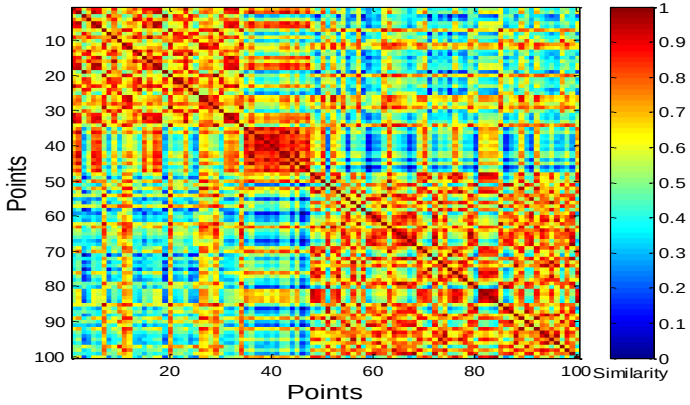
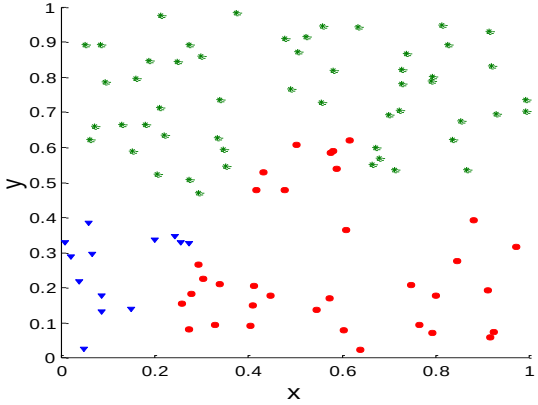
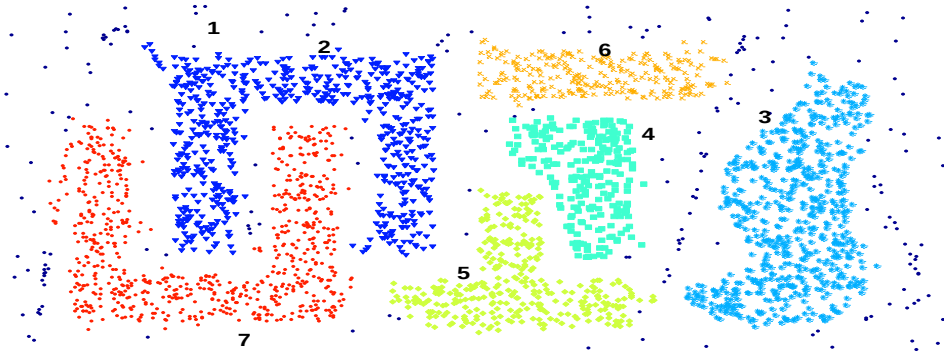
- Clusters in random data are not so crisp

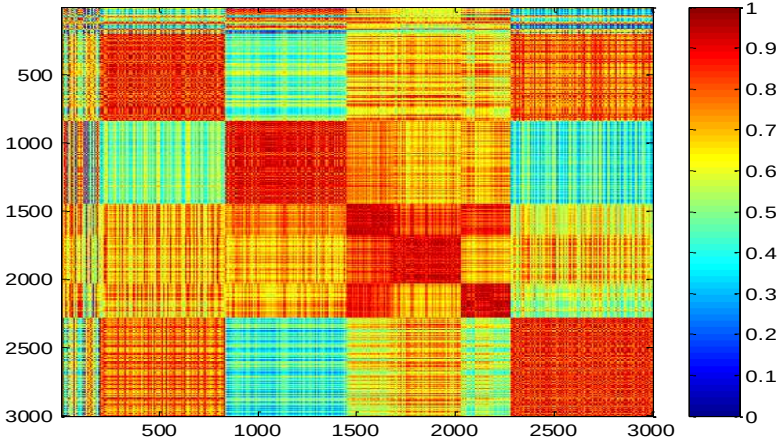




Complete Link

Using Similarity Matrix for Cluster Validation

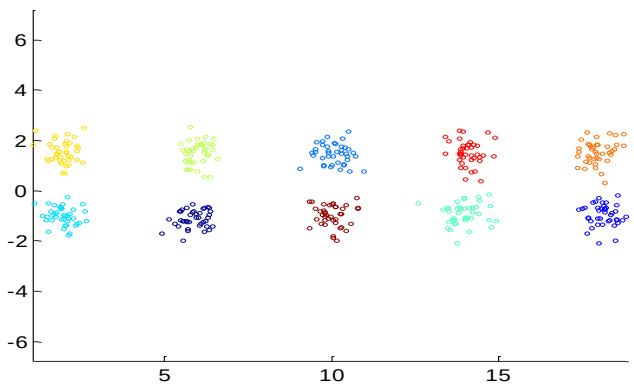


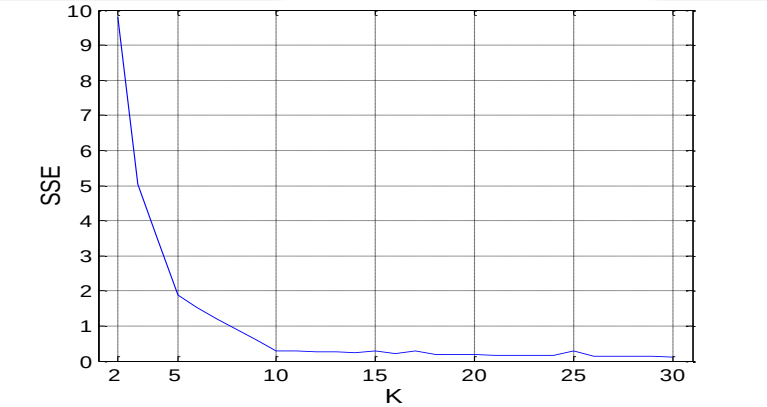


DBSCAN

Internal Measures: SSE

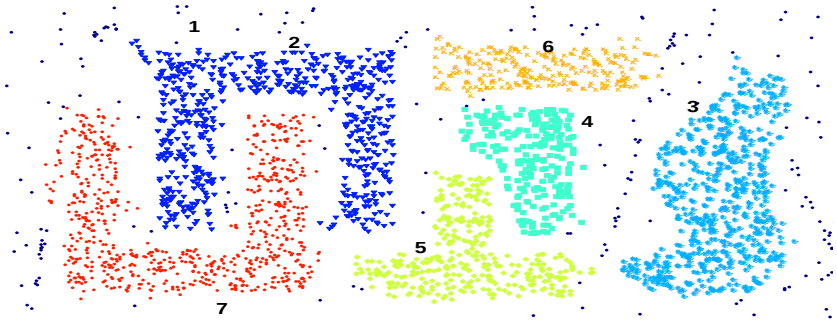
- Clusters in more complicated figures aren't well separated
- Internal Index: Used to measure the goodness of a clustering structure without respect to external information
  - SSE
- SSE is good for comparing two clusterings or two clusters (average SSE).
- Can also be used to estimate the number of clusters

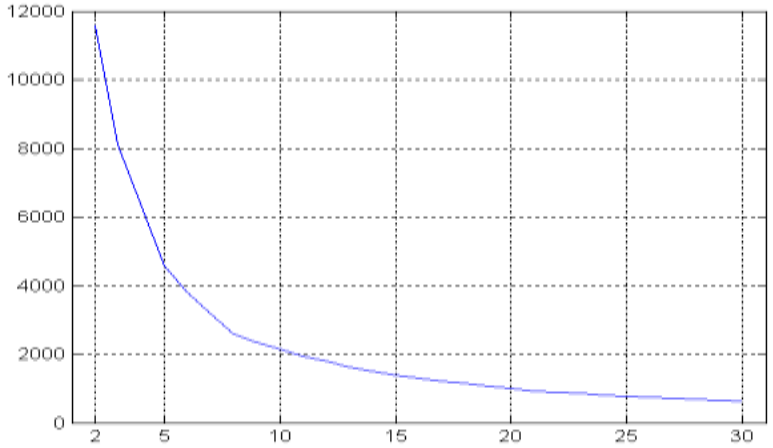




Internal Measures: SSE

- SSE curve for a more complicated data set





SSE of clusters found using K-means

Framework for Cluster Validity

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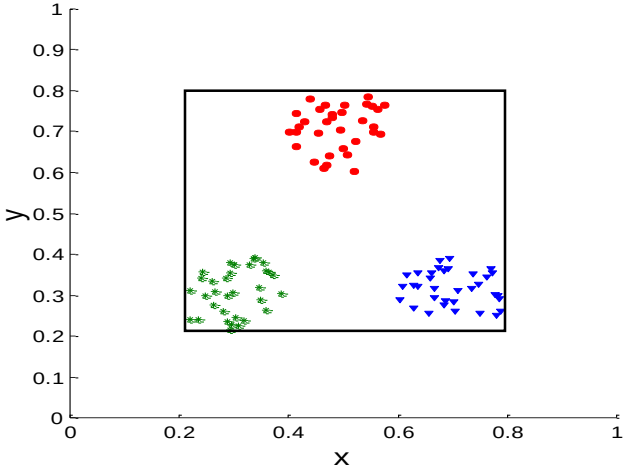
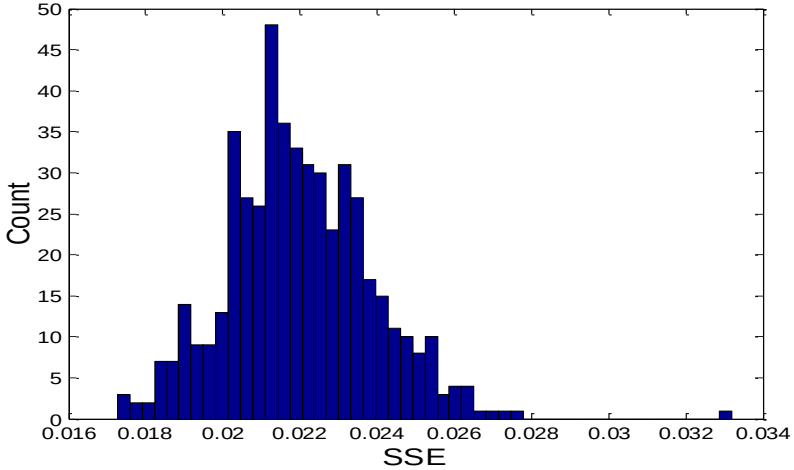
- Need a framework to interpret any measure.
  - For example, if our measure of evaluation has the value, 10, is that good, fair, or poor?
- Statistics provide a framework for cluster validity
  - The more "atypical" a clustering result is, the more likely it represents valid structure in the data
  - Can compare the values of an index that result from random data or clusterings to those of a clustering result.
    - If the value of the index is unlikely, then the cluster results are valid
  - These approaches are more complicated and harder to understand.
- For comparing the results of two different sets of cluster analyses, a framework is less necessary.
  - However, there is the question of whether the difference between two index values is significant

Statistical Framework for SSE

- Example
  - Compare SSE of 0.005 against three clusters in random data
  - Histogram shows SSE of three clusters in 500 sets of random data points of size 100 distributed over the range 0.2 – 0.8 for x and y values





Statistical Framework for Correlation

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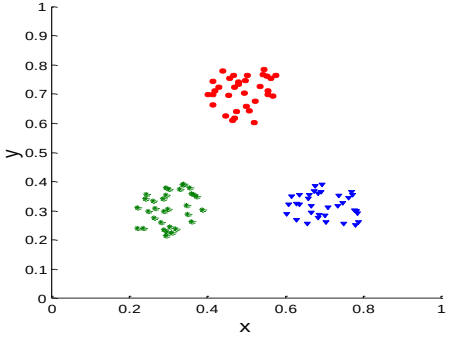
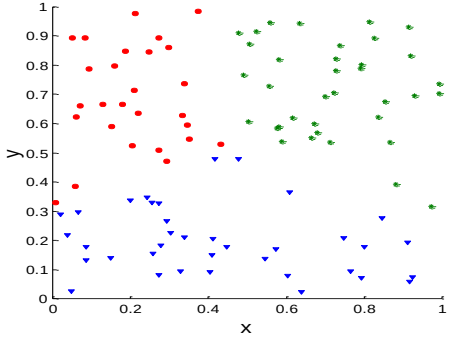
- Correlation of incidence and proximity matrices for the K-means clusterings of the following two data sets.



Corr = -0.9235



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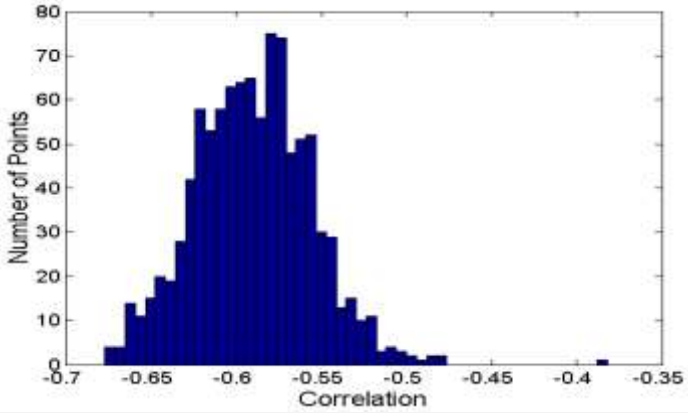


- Cluster Cohesion: Measures how closely related are objects in a cluster.
- Cluster Separation: Measure how distinct or well-separated a cluster is from other clusters
- Example: Squared Error
  - Cohesion is measured by the within cluster sum of squares (SSE)- Measures how tightly points are grouped around their cluster centroid. Lower WSS → clusters are compact. Mi is the centroid

$$WSS = \sum_i \sum_{x \in C_i} (x - m_i)^2$$

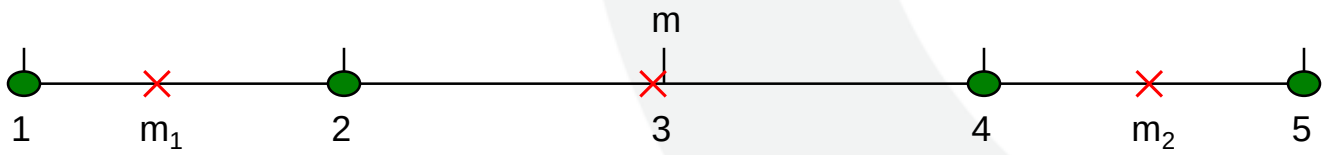
  - Separation is measured by the between cluster sum of squares- Measures how far apart cluster centroids are from the overall mean. Higher BSS → clusters are well-separated.
    - Where $|C_i|$ is the size of cluster I and m is global mean

$$BSS = \sum_i |C_i|(m - m_i)^2$$

Internal Measures: Cohesion and Separation

- Example: SSE
  - BSS + WSS = constant

Increasing k- WSS decreases-tight clusters
Increasing k- BSS increases – clusters far apart



K=1 cluster:

$$WSS = (1-3)^2 + (2-3)^2 + (4-3)^2 + (5-3)^2 = 10$$

$$BSS = 4 \times (3-3)^2 = 0$$

$$Total = 10 + 0 = 10$$

K=2 clusters:

$$WSS = (1-1.5)^2 + (2-1.5)^2 + (4-4.5)^2 + (5-4.5)^2 = 1$$

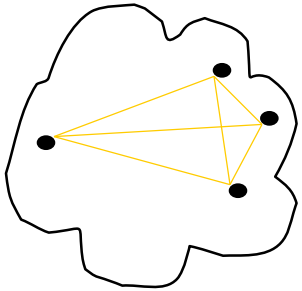
$$BSS = 2 \times (3-1.5)^2 + 2 \times (4.5-3)^2 = 9$$

$$Total = 1 + 9 = 10$$

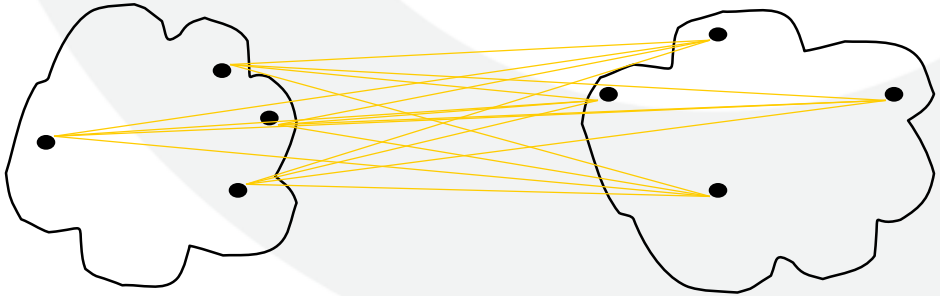
Clusters:
$C_1 = \{1, 2\}$,mean $\mu_1 = 1.5$
$C_2 = \{4, 5\}$,mean $\mu_2 = 4.5$

Internal Measures: Cohesion and Separation

- A proximity graph based approach can also be used for cohesion and separation.
  - Cluster cohesion is the sum of the weight of all links within a cluster.
  - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



cohesion



separation

Internal Measures: Silhouette Coefficient

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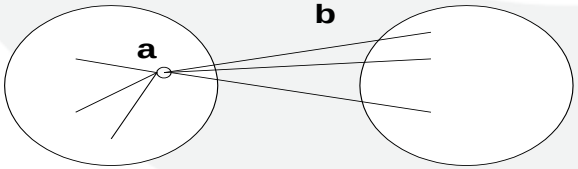
- Silhouette Coefficient combine ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings
- For an individual point, *i*
  - Calculate ***a*** = average distance of *i* to the points in its cluster
  - Calculate ***b*** = min (average distance of *i* to points in another cluster)
  - The silhouette coefficient for a point is then given by

    $s = 1 - a/b$ if $a < b$, (or $s = b/a - 1$ if $a \geq b$, not the usual case)

  - Typically between 0 and 1.
  - The closer to 1 the better.

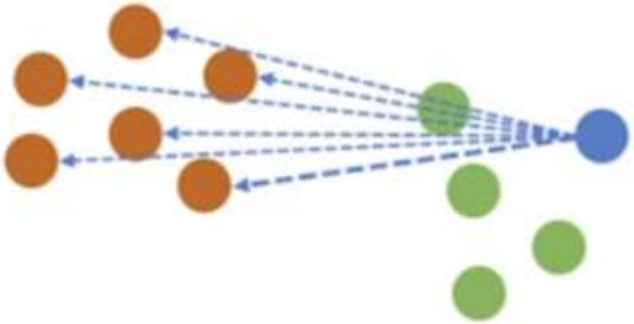


- Can calculate the Average Silhouette width for a cluster or a clustering

Internal Measures: Silhouette Coefficient

a(i): avg distance between i and all other datapoints within cluster

a(i)



b(i): avg distance between i and all other datapoints outside/neighboring cluster

b(i)

Internal Measures: Silhouette Coefficient

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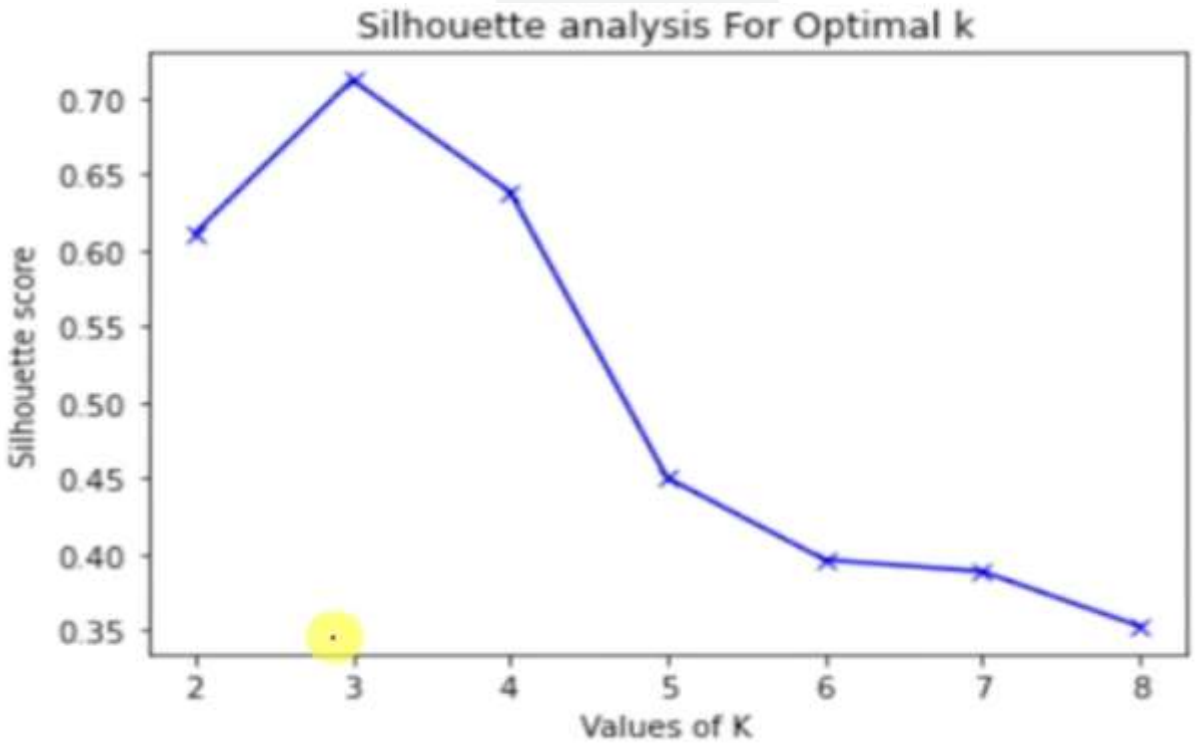

External Measures of Cluster Validity: Entropy and Purity

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Table 5.9. K-means Clustering Results for LA Document Data Set

Cluster	Entertainment	Financial	Foreign	Metro	National	Sports	Entropy	Purity
1	3	5	40	506	96	27	1.2270	0.7474
2	4	7	280	29	39	2	1.1472	0.7756
3	1	1	1	7	4	671	0.1813	0.9796
4	10	162	3	119	73	2	1.7487	0.4390
5	331	22	5	70	13	23	1.3976	0.7134
6	5	358	12	212	48	13	1.5523	0.5525
Total	354	555	341	943	273	738	1.1450	0.7203

entropy For each cluster, the class distribution of the data is calculated first, i.e., for cluster j we compute p_{ij}, the 'probability' that a member of cluster j belongs to class i as follows: $p_{ij} = m_{ij}/m_j$, where m_j is the number of values in cluster j and m_{ij} is the number of values of class i in cluster j. Then using this class distribution, the entropy of each cluster j is calculated using the standard formula $e_j = \sum_{i=1}^{L} p_{ij} \log_2 p_{ij}$, where the L is the number of classes. The total entropy for a set of clusters is calculated as the sum of the entropies of each cluster weighted by the size of each cluster, i.e., $e = \sum_{i=1}^{K} \frac{m_i}{m} e_j$, where m_j is the size of cluster j, K is the number of clusters, and m is the total number of data points.

purity Using the terminology derived for entropy, the purity of cluster j, is given by $purity_j = \max p_{ij}$ and the overall purity of a clustering by $purity = \sum_{i=1}^{K} \frac{m_i}{m} purity_j$.

Final Comment on Cluster Validity

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"The validation of clustering structures is the most difficult and frustrating part of cluster analysis.

Without a strong effort in this direction, cluster analysis will remain a black art accessible only to those true believers who have experience and great courage."

Algorithms for Clustering Data, Jain and Dubes

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[1] A. B. Amjoud and M. Amrouch, "Object Detection Using Deep Learning, CNNs and Vision Transformers: A Review," in IEEE Access, vol. 11, pp. 35479-35516, 2023, doi: 10.1109/ACCESS.2023.3266093.